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AI-driven bioactive peptide discovery of next-generation metabolic biotherapeutics

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ABSTRACT

Metabolic diseases, including obesity and type 2 diabetes, pose a significant global health burden, demanding innovative therapeutic solutions. Traditional drug discovery is often slow and costly, struggling with the complex nature of these disorders. Bioactive peptides offer a promising alternative due characterized by their specificity, low toxicity, and diverse mechanisms. However, challenges in their screening, stability, and target identification have limited their clinical use. Artificial intelligence (AI) and machine learning (ML) are now revolutionizing peptide discovery. These technologies enable rapid prediction, *de novo* design, and optimization of bioactive sequences. This review critically evaluates AI's role in identifying and developing peptides for metabolic disease pathways. We examine key computational methods, including sequence-based features, advanced deep learning models (CNNs, LSTMs, Transformers), and generative approaches. The manuscript also covers essential datasets, validation frameworks, and illustrative case studies. We explore the integration of molecular dynamics, network pharmacology, and reinforcement learning for advanced peptide engineering. Despite significant progress, challenges persist, such as data heterogeneity, model generalizability, and the gap between *in silico* predictions and experimental validation. Looking ahead, we highlight future opportunities, including multi-omics integration, explainable AI, the discovery of microbiome-derived peptides, and synthetic biology-driven design. This review underscores AI's transformative potential in advancing peptide-based interventions for metabolic diseases, offering a roadmap for novel, targeted, and preventive therapies.

1. Introduction

The global burden of metabolic diseases has reached epidemic proportions, with conditions such as obesity, type 2 diabetes mellitus (T2DM), non-alcoholic fatty liver disease (NAFLD), and metabolic syndrome affecting hundreds of millions of individuals worldwide (Henning & Greene, 2023; Kalligeros et al., 2024; Katsi et al., 2023; La Merrill et al., 2025; Pigeot & Ahrens, 2025). These disorders are not only highly prevalent but also deeply interconnected, sharing common pathophysiological mechanisms including insulin resistance, chronic low-grade inflammation, dyslipidemia, and altered gut microbiota composition (Hamadou et al., 2022; Katsi et al., 2023; La Merrill et al., 2025). Despite advances in pharmacotherapy and lifestyle interventions, the current therapeutic arsenal remains suboptimal for many patients, often failing to address the root causes or provide long-term prevention. This underscores an urgent need for innovative, targeted, and sustainable strategies aimed at both treatment and prevention.

In this context, bioactive peptides have emerged as promising candidates due to their diverse biological functions and relatively favorable safety profiles (Mamoudou, Başaran, et al., 2024; Wayal et al., 2025). Derived from food proteins, endogenous precursors, or microbial sources, these short amino acid sequences exert a wide range of physiological

Abbreviations: AI, Artificial Intelligence; CNN, Convolutional Neural Network; ESM, Evolutionary Scale Modeling; GAN, Generative Adversarial Network; LIME, Local Interpretable Model-Agnostic Explanations; LSTM, Long Short-Term Memory; ML, Machine Learning; ProtBERT, Protein BERT; RF, Random Forest; RL, Reinforcement Learning; RNN, Recurrent Neural Network; SHAP, SHapley Additive exPlanations; SVM, Support Vector Machine; T2DM, Type 2 Diabetes Mellitus; VAE, Variational Autoencoder; XAI, Explainable Artificial Intelligence; XGBoost, Extreme Gradient Boosting.

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effects, such as modulating glucose homeostasis, suppressing appetite, reducing inflammation, and improving lipid metabolism, that are directly relevant to metabolic health (Mamoudou, Obadias, et al., 2024; Mamoudou & Mune, 2024; Mansoori et al., 2025). Unlike traditional small-molecule drugs or large biologics, bioactive peptides offer a unique balance between specificity and tolerability, making them attractive candidates for functional foods, nutraceuticals, and even pharmaceutical agents (Antony & Vijayan, 2021; Hamadou, 2025; Mamoudou & Mune, 2024).

However, the discovery and development of effective bioactive peptides remain challenging. Conventional methods rely heavily on labor-intensive biochemical assays, animal models, and time-consuming structure-activity relationship studies. These approaches are often limited by low throughput, high costs, and poor scalability (Hamadou et al., 2025). Moreover, predicting key properties such as bioavailability, enzymatic stability, target selectivity, and ADMET (absorption, distribution, metabolism, excretion, and toxicity) profiles remains a significant hurdle in translating promising peptides from bench to bedside (Kifle et al., 2020; Lanka et al., 2023; Mamoudou et al., 2025).

This is where artificial intelligence (AI) and machine learning (ML) are revolutionizing the field. By leveraging vast sequence data, structural information, and biological annotations, AI-based tools can accelerate the identification, prediction of functional properties, and rational design of novel bioactive peptides (Gangwal & Lavecchia, 2024; Ivanenkov et al., 2023; Kamya et al., 2024; Saini et al., 2025). AI models, ranging from deep learning architectures recognizing sequence-activity patterns to generative algorithms designing entirely new peptides, are fundamentally reshaping peptide discovery.

Recent years have indeed seen a surge in computational pipelines integrating AI with experimental validation (Gangwal & Lavecchia, 2024; Ivanenkov et al., 2023; Kamya et al., 2024; Saini et al., 2025). This allows for *in silico* screening of thousands of candidates, optimizing resource allocation for subsequent wet-lab testing. These AI models are increasingly trained on curated public datasets, enhanced by sophisticated feature engineering, and rigorously validated. Furthermore, the integration of Natural Language Processing (NLP)-inspired methods, such as transformer-based models (V. Kumar et al., 2025), now permits a nuanced understanding of peptide sequence-function relationships, mirroring how linguistic meaning is interpreted (Cai et al., 2025; V. Kumar et al., 2025; Mullowney et al., 2023).

Despite these advancements, several challenges persist. Data heterogeneity, model overfitting, lack of standardization, and the difficulty of extrapolating findings across species or populations remain critical issues (Y. Lin et al., 2024; Sheikh et al., 2013). Furthermore, while AI excels at pattern recognition and prediction, bridging the gap between computational output and biological plausibility requires careful interpretation and rigorous experimental follow-up (Cai et al., 2025; Lavecchia, 2025; Saini et al., 2025; Yingngam & Sethabouppha, 2024). Thus, successful implementation of AI in this domain demands close collaboration between computational scientists, molecular biologists, nutritionists, and clinicians.

The objective of this comprehensive review is to critically evaluate the current state of AI applications in the discovery of bioactive peptides for the prevention and management of metabolic diseases. While existing reviews often broadly cover AI in drug discovery, this manuscript uniquely focuses on the specific challenges and opportunities within the peptide space, particularly addressing the complex landscape of metabolic disorders, and integrates discussions on emerging translational aspects like microbiome-derived peptides and intellectual property trends. We explore the methodologies underpinning AI-driven peptide discovery, assess the quality and scope of available datasets, analyze the performance of various machine learning and deep learning architectures, and highlight notable case studies where AI has led to tangible discoveries. Additionally, we discuss the limitations of existing approaches, outline emerging opportunities for innovation, and propose future directions for research aimed at enhancing the translational potential of AI in this rapidly evolving field.

By synthesizing knowledge across disciplines, from computational biology and systems medicine to nutritional science and drug design, this review aims to serve as a reference for researchers seeking to harness the power of artificial intelligence in the quest for novel, effective, and safe peptide-based interventions against metabolic diseases.

2. Methodology

This review was conducted following a systematic and transparent approach to ensure the inclusion of high-quality, relevant, and up-to-date literature on the application of artificial intelligence (AI) in the discovery of bioactive peptides for the prevention and treatment of metabolic diseases. The methodology adhered to best practices in evidence-based scientific reviewing while allowing for flexibility to capture the rapidly evolving nature of computational biology and AI-driven drug discovery.

2.1. Literature search strategy

A comprehensive search was performed across multiple electronic databases, including PubMed, Scopus, Web of Science, IEEE Xplore, and Google Scholar, using a combination of controlled vocabulary and free-text terms related to key domains: *artificial intelligence, machine learning, bioactive peptides, metabolic diseases, in silico screening, peptide design*, and *computational drug discovery*. Boolean operators were used to combine keywords and synonyms effectively, ensuring broad coverage of relevant publications. Filters included peer-reviewed articles published in English between January 2000 and the present, with preference given to studies reporting empirical results, predictive models, or novel methodologies (Helbach et al., 2023; Moher et al., 2010).

2.2. Inclusion and exclusion criteria

To maintain scientific relevance and focus, the following inclusion criteria were applied:

- Original research articles, reviews, or technical reports involving AI/ML techniques applied to peptide discovery.
- Studies focusing on bioactive peptides with demonstrated or predicted roles in modulating metabolic disease pathways.
- Papers describing computational pipelines, predictive models, or *in silico* tools validated either computationally or experimentally.
- Works that explicitly mention metabolic conditions such as obesity, type 2 diabetes, NAFLD, insulin resistance, or metabolic syndrome.

Exclusion criteria encompassed:

- Non-English publications.
- Abstract-only proceedings without full-text access.
- Editorials and commentaries.
- Studies not clearly linking AI methods to peptide discovery or metabolic outcomes.

2.3. Study selection and data extraction

The initial search yielded several hundred records, which were first screened by title and abstract for relevance. Duplicate entries were removed using reference management software (Mendeley Reference v.2.122.1). Full-text articles deemed potentially eligible were then assessed independently by two reviewers based on predefined inclusion criteria. Discrepancies were resolved through discussion or consultation with a third reviewer. Data extraction focused on key elements including:

- Study objectives and scope
- AI/ML techniques employed
- Peptide sources and types
- Disease targets and biological mechanisms
- Validation methods (*in vitro, in vivo, in silico,* clinical)
- Model performance metrics and limitations

2.4. Quality assessment and risk of bias

Due to the heterogeneous nature of studies included, ranging from computational modeling to experimental validation, a formal risk-of-bias assessment tool was adapted from existing frameworks used in systematic reviews of diagnostic or predictive models. Each study was evaluated for methodological soundness, data quality, model interpretability, and reproducibility. Particular attention was paid to the robustness of training datasets, cross-validation strategies, and whether independent testing was performed.

2.5. Synthesis and narrative integration

Findings were synthesized thematically and organized according to the major conceptual areas outlined in the review structure: background on metabolic diseases, bioactive peptides, AI methodologies, applications, challenges, and future directions. Emphasis was placed on identifying trends, technological advancements, and gaps in the current literature. Where applicable, representative case studies and illustrative examples were selected to highlight innovative uses of AI in the field.

2.6. Limitations of the review approach

Despite efforts to ensure comprehensiveness, some limitations are acknowledged. First, the exclusion of non-English publications may have introduced language bias. Second, due to the rapid pace of AI development, some recent preprints or unpublished studies may not be fully represented. Third, the lack of standardized reporting formats across AI-driven peptide discovery studies made direct comparisons challenging. However, wherever possible, we prioritized peer-reviewed, high-impact publications to ensure scientific credibility.

3. Metabolic diseases: global burden and therapeutic challenges

Metabolic diseases represent a major public health crisis of the 21st century, with their prevalence rising in parallel with global urbanization, sedentary lifestyles, and dietary shifts toward energy-dense, nutrient-poor foods (Hamadou et al., 2022; Henning & Greene, 2023; La Merrill et al., 2025). Among these, obesity, type 2 diabetes mellitus (T2DM), non-alcoholic fatty liver disease (NAFLD), insulin resistance, and hypertension stand out as key components of metabolic syndrome, a constellation of interrelated conditions that significantly elevate the risk of cardiovascular disease, stroke, and chronic kidney disease (Alves et al., 2019; Chiș et al., 2023; Pigeot & Ahrens, 2025). According to the World Health Organization (WHO), over 650 million adults were classified as obese in 2022, while more than 420 million individuals suffer from diabetes globally, the vast majority of whom have T2DM (La Merrill et al., 2025; Pigeot & Ahrens, 2025; WHO, 2015). NAFLD affects approximately one-quarter of the global population, often coexisting with obesity and insulin resistance. Hypertension, meanwhile, remains the leading modifiable risk factor for premature mortality worldwide, contributing to nearly 11 million deaths annually (Hisamatsu & Miura, 2021; La Merrill et al., 2025; Pigeot & Ahrens, 2025).

The pathophysiology underlying these disorders is complex and multifactorial, involving dysregulated glucose and lipid metabolism, chronic low-grade inflammation, oxidative stress, mitochondrial dysfunction, and alterations in gut microbiota. Insulin resistance, defined as the diminished ability of tissues such as skeletal muscle, liver, and adipose tissue to respond to insulin, is a central feature linking many of these conditions (La Merrill et al., 2025; Luka et al., 2024; Pigeot & Ahrens, 2025). It drives compensatory hyperinsulinemia, which in turn exacerbates hepatic gluconeogenesis, lipogenesis, and systemic inflammation, ultimately promoting the development of T2DM and NAFLD (La Merrill et al., 2025; Pigeot & Ahrens, 2025).

Despite the availability of several pharmacological interventions (metformin, GLP-1 receptor agonists, SGLT2 inhibitors, thiazolidinediones, statins, and antihypertensive agents) the management of metabolic diseases remains suboptimal for many patients (Aleixandre & Miguel, 2011; Nganso Ditchou et al., 2024; Sarker et al., 2024; S. Zhang et al., 2016). These therapies, while effective in some cases, are often associated with limitations such as variable efficacy, undesirable side effects, high costs, and poor long-term adherence. For instance, metformin, the first-line treatment for T2DM, can cause gastrointestinal discomfort and is contraindicated in patients with renal impairment. GLP-1 receptor agonists, though increasingly recognized for their weight-lowering and cardioprotective effects, are expensive and require injectable administration, limiting accessibility in resource-constrained settings (Abubakar et al., 2021; Leutcha et al., 2025; Mamoudou et al., 2025; Nnemolisa et al., 2024; Sarker et al., 2024). Similarly, lifestyle modifications such as diet and exercise remain underutilized or unsustainable for many individuals due to socioeconomic barriers, lack of motivation, or biological resistance to weight loss.

Moreover, most current treatments are primarily aimed at symptom management rather than addressing the root causes of metabolic dysfunction (Henning & Greene, 2023; La Merrill et al., 2025). There is a growing recognition that effective prevention and early intervention strategies are essential to halt disease progression before irreversible damage occurs (Y. Lin et al., 2024; Nganso Ditchou et al., 2020). This has spurred interest in identifying novel therapeutic modalities that can target the molecular and cellular mechanisms underlying metabolic dysregulation with greater precision and fewer side effects.

In this context, bioactive peptides have garnered significant attention as promising candidates for both therapeutic and preventive applications (Hamadou, 2025; Mamoudou & Mune, 2024). Their ability to interact with specific molecular targets involved in metabolic regulation (incretin receptors, G protein-coupled receptors (GPCRs), nuclear hormone receptors, and enzymes like dipeptidyl peptidase-4 (DPP-4)) makes them attractive tools for modulating glucose homeostasis, lipid metabolism, appetite control, and inflammatory pathways (Habib et al., 2025; Hamadou, 2025; La Merrill et al., 2025; Mamoudou et al., 2025; Mune et al., 2018). Unlike conventional drugs, bioactive peptides often exhibit high specificity, reduced toxicity, and pleiotropic effects, making them well-suited for multi-targeted approaches in complex diseases (Mamoudou, Başaran, et al., 2024; Mora et al., 2019).

However, the discovery and development of effective bioactive peptides face substantial challenges, including limited throughput in traditional screening methods, poor bioavailability, rapid enzymatic degradation, and difficulties in predicting functional properties from sequence data alone (Antony & Vijayan, 2021; Mamoudou et al., 2025; Mamoudou & Mune, 2024). This is where artificial intelligence (AI) and machine learning (ML) are proving transformative, offering new avenues to accelerate peptide identification, optimize design, and enhance translational potential (Abraham et al., 2021; Cè et al., 2022; Y. Yang & Cheng, 2025).

Thus, the urgent need for innovative, targeted, and preventive interventions against metabolic diseases has never been clearer. The integration of AI into peptide research not only promises to overcome longstanding bottlenecks but also opens the door to personalized, mechanism-based therapies tailored to individual metabolic profiles. In the following sections, we will explore how computational approaches are reshaping the landscape of bioactive peptide discovery and paving the way for next-generation metabolic therapeutics.

4. Bioactive peptides as emerging therapeutics

Bioactive peptides represent a rapidly expanding class of functional biomolecules with significant potential in the prevention and treatment of metabolic diseases (Hamadou, 2025; Mesias et al., 2025; Mora et al., 2019; Tagliamonte et al., 2025). These short sequences of amino acids, typically comprising 3 to 30 residues, are capable of exerting a wide range of physiological effects beyond their nutritional value (Mamoudou & Mune, 2024; Mora et al., 2019; Tagliamonte et al., 2025). Unlike bulk dietary proteins, bioactive peptides are often cryptic within the parent protein and become biologically active only after enzymatic hydrolysis during digestion or food processing (Mamoudou, Başaran, et al., 2024; Mamoudou & Mune, 2024; Marcone et al., 2017; Mora et al., 2019). Their ability to interact selectively with molecular targets, modulate key signaling pathways (Hamadou, 2025; Mamoudou & Mune, 2024), and exhibit minimal off-target toxicity positions them as promising candidates for next-generation therapeutics.

4.1. Definition and origin of bioactive peptides

Bioactive peptides can be broadly classified based on their origin into three major categories (Habib et al., 2025; Marcone et al., 2017; Mora et al., 2019; Wu, 2016):

- Dietary-derived peptides originate from food proteins such as casein, whey, soy, fish, and plant-based sources. They are released during gastrointestinal digestion or fermentation by proteolytic enzymes or microbial fermentation. Peptides derived from milk casein have been extensively studied for their antihypertensive properties (Bayang et al., 2025; Habib et al., 2025; Himeda et al., 2022; Marcone et al., 2017; Mora et al., 2019; Wu, 2016).
- Endogenous peptides are naturally produced within the body through post-translational modification of precursor proteins. Examples include incretin hormones such as glucagon-like peptide-1 (GLP-1) and glucose-dependent insulinotropic polypeptide (GIP), which play central roles in glucose homeostasis. Neuropeptides like ghrelin and leptin also regulate appetite and energy balance, making them relevant in obesity and metabolic syndrome (Marcone et al., 2017; Mesias et al., 2025; Mora et al., 2019; Stavrakov et al., 2021; G. Wu, 2016).
- Microbiota-derived peptides originate from gut microbial metabolism of host or dietary proteins. The gut microbiome acts as an extensive biochemical factory, generating bioactive metabolites that can influence systemic metabolism via enteroendocrine signaling, immune modulation, and neural pathways. Short-chain fatty acid (SCFA)-producing bacteria generate signaling molecules that affect host metabolism and inflammation (Kasubuchi et al., 2015; Marcone et al., 2017; Mora et al., 2019; Stavrakov et al., 2021; G. Wu, 2016; Yutharaksanukul et al., 2024).

These distinct origins highlight the diverse sources of bioactive peptides and underscore the importance of considering both exogenous and endogenous contributions when evaluating their therapeutic potential.

4.2. Mechanisms of action

The biological activities of bioactive peptides are mediated through multiple mechanisms, often involving direct interaction with cellular receptors, enzyme inhibition, or modulation of signaling cascades (Fig. 1). Collectively, these mechanisms illustrate the multifunctional nature of bioactive peptides, enabling them to act at multiple levels, hormonal, enzymatic, immunological, and microbial, to address the complex pathophysiology of metabolic disorders (Habib et al., 2025; Hamadou, 2025; Tagliamonte et al., 2025).

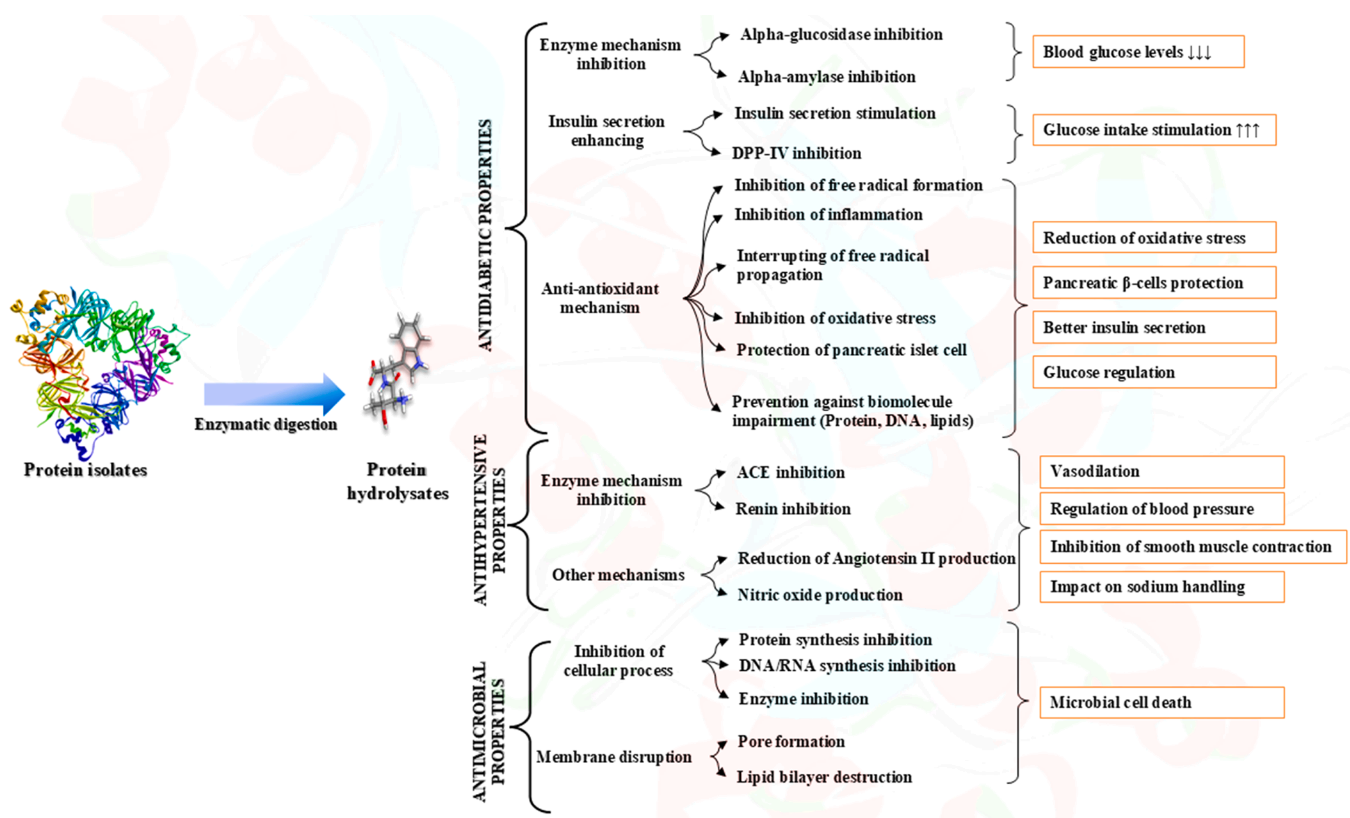

Fig. 1. Mechanisms of action of bioactive peptides (Mamoudou & Mune, 2024).

4.3. Advantages over small molecules and biologics

Bioactive peptides represent a promising class of therapeutic agents for the management of metabolic diseases (Fig. 1), offering distinct advantages over conventional small-molecule drugs and large biologics such as monoclonal antibodies and recombinant proteins (Habib et al., 2025; Mesias et al., 2025; Tagliamonte et al., 2025). One of their most notable attributes is their high specificity and target selectivity. Structurally resembling endogenous ligands, bioactive peptides often demonstrate a strong affinity for specific receptors or enzymes. This inherent selectivity minimizes the likelihood of off-target interactions, which are frequently observed with small-molecule drugs that tend to bind non-specifically to various cellular components (Amang à Ngnoung et al., 2023; Asledottir et al., 2023; Mamoudou, Başaran, et al., 2024; Mune et al., 2018; Oscar Ditchou Nganso, Sidjui, et al., 2020). Consequently, peptides can achieve therapeutic efficacy with reduced unintended effects, making them particularly appealing for precision medicine applications.

In addition to their specificity, bioactive peptides exhibit a favorable safety profile, largely attributed to their natural composition of amino acids (Habib et al., 2025; Mamoudou & Mune, 2024; Mesias et al., 2025; Tagliamonte et al., 2025). These peptides are typically metabolized into innocuous fragments by endogenous peptidases, thereby reducing the risk of systemic toxicity. This stands in contrast to many synthetic drugs, which may carry hepatotoxic or nephrotoxic liabilities. Furthermore, peptides possess multifunctional properties, enabling them to interact with multiple biological targets simultaneously, a phenomenon referred to as polypharmacology (Habib et al., 2025; Mamoudou & Mune, 2024; Mesias et al., 2025; Tagliamonte et al., 2025). For instance, certain peptides have been shown to regulate glucose levels, mitigate inflammation, and enhance satiety concurrently (Hamadou et al., 2020; Rivera-Jiménez et al., 2022; Tagliamonte et al., 2025; Weber et al., 2023). This ability to address multiple facets of complex metabolic syndromes underscores their potential to provide a more comprehensive and integrative therapeutic approach. Another key advantage of bioactive peptides is their relatively low immunogenic potential when compared to large biologics (Bellaver & Kempka, 2023; Habib et al., 2025; Tagliamonte et al., 2025). While large biologics often elicit immune responses due to their structural complexity, peptides are generally less likely to provoke such reactions (Bellaver & Kempka, 2023; Habib et al., 2025; Tagliamonte et al., 2025).

However, it is important to note that this benefit may be compromised if peptides are chemically modified or delivered using adjuvants or carriers that enhance immune recognition. Additionally, the dietary integration of bioactive peptides presents a unique opportunity for preventive healthcare. Many of these peptides are naturally derived from food sources, allowing them to be incorporated into functional foods or nutraceuticals. This non-invasive strategy holds significant promise for at-risk populations, offering a means to promote metabolic health without the need for medical intervention.

Despite these compelling advantages, the clinical translation of bioactive peptides is not without challenges. A major limitation is their poor oral bioavailability, primarily due to rapid enzymatic degradation in the gastrointestinal tract. Moreover, peptides often exhibit a short half-life and face difficulties in crossing biological barriers, such as the intestinal epithelium or the blood-brain barrier. These obstacles have spurred significant research efforts aimed at overcoming these limitations. Notably, advancements in artificial intelligence and computational modeling have emerged as powerful tools in this endeavor, facilitating the design of peptide-based therapeutics with enhanced stability, bioavailability, and delivery efficiency.

5. Challenges in traditional peptide discovery

Despite the therapeutic promise of bioactive peptides, their discovery and development have historically relied on conventional, labor-intensive methodologies that are both time-consuming and resource-demanding. These traditional approaches typically involve isolating peptides from natural sources, such as food proteins or tissue extracts, followed by biochemical assays to evaluate biological activity. While such methods have led to the identification of several important peptides, they are increasingly recognized as inefficient and inadequate for addressing the complexity and scale required in modern drug discovery.

5.1. Labor-intensive screening processes

Traditional peptide discovery often begins with enzymatic hydrolysis of precursor proteins under controlled conditions to release potential bioactive fragments (Mamoudou & Mune, 2024; Su et al., 2024). These hydrolysates are then subjected to fractionation techniques, including gel filtration chromatography, reverse-phase high-performance liquid chromatography (RP-HPLC), and mass spectrometry (MS)-based sequencing (Admassu et al., 2018; Mune et al., 2018; J. Zhang et al., 2020). Each step requires specialized equipment and skilled personnel, making the process slow and prone to variability. Moreover, the sheer number of possible cleavage sites and resulting peptide sequences necessitates extensive downstream testing, significantly prolonging the timeline from initial isolation to functional characterization (L. He et al., 2024; Mamoudou, Başaran, et al., 2024).

5.2. Limited throughput and scalability

One of the most significant limitations of classical screening methods is their low throughput. Conventional assays such as enzyme inhibition assays, cell-based receptor binding studies, or animal models, are not amenable to high-throughput formats (Elisha et al., 2025; Mkabayi et al., 2024; Oscar Ditchou Nganso, Marthe Satchet Tchana, et al., 2020). As a result, only a small subset of potentially active peptides can be tested in detail, leading to a biased selection toward already known or easily detectable sequences. This bottleneck becomes even more pronounced when attempting to explore combinatorial libraries or synthetic analogs designed to enhance potency or stability.

5.3. Poor prediction of admet properties

Traditional drug discovery pipelines exhibit a notable limitation in their capacity to accurately predict ADMET (absorption, distribution, metabolism, excretion, and toxicity) properties during the early stages of development (Guan et al., 2019). This limitation frequently results in the unsuccessful progression of numerous promising peptides in preclinical or clinical trials, mainly attributable to adverse pharmacokinetic characteristics.

Absorption is often impaired in peptide-based therapeutics. Oral delivery, an advantageous method for enhancing patient compliance, frequently faces challenges due to inadequate membrane permeability and vulnerability to degradation by gastrointestinal proteases (Tagliamonte et al., 2025; Wang et al., 2020). The bioavailability of peptides administered via this route is significantly limited by these factors. Furthermore, distribution presents a significant challenge. Peptides often demonstrate restricted tissue penetration and may experience swift renal clearance, which diminishes their capacity to effectively reach target sites and maintain therapeutic concentrations over time (Larder et al., 2023).

Many peptides are rapidly degraded by endogenous peptidases, resulting in a short half-life that diminishes their therapeutic efficacy (Tagliamonte et al., 2025; Wang et al., 2020). Toxicity continues to be a concern, as off-target effects or immunogenic responses may only manifest following extensive animal or human testing (Gupta et al., 2013). These issues present significant challenges due to their detection difficulties with conventional biochemical assays in the early developmental stages.

Due to these complexities, researchers frequently engage in empirical optimization via iterative cycles of synthesis and testing. This approach seeks to refine peptide candidates but is inherently costly, inefficient, and often unsuccessful. The dependence on trial-and-error approaches highlights the critical necessity for enhanced predictive and mechanistic tools to assess ADMET properties at earlier stages of the discovery process. Addressing these gaps may substantially improve the success rates of peptide-based therapeutics and facilitate their transition from laboratory to clinical application.

Given these challenges, innovative strategies, including computational modeling, artificial intelligence, and advanced screening platforms, are being investigated to address the limitations of traditional pipelines. These methods show potential for enhancing the early prediction of ADMET properties and facilitating the rational design of peptides with optimized pharmacokinetic and safety profiles.

5.4. High failure rates in clinical trials

These limitations ultimately lead to elevated attrition rates in clinical development. Data from the industry indicate that fewer than 10 % of peptides that enter preclinical testing progress to clinical trials, with only a small percentage receiving regulatory approval (Duffuler et al., 2022). Failures are frequently associated with several persistent issues candidates demonstrating efficacy in preclinical studies often do not replicate results in human trials, unexpected safety concerns arise, inadequate bioavailability requires invasive delivery methods such as injections, and challenges in manufacturing complicate the scaling of peptide production (Duffuler et al., 2022).

The challenges highlight the urgent requirement for enhanced predictive, scalable, and efficient discovery methods, which has led to the increased utilization of computational tools and artificial intelligence in peptide research.

6. The rise of artificial intelligence in biomedical research

The advent of artificial intelligence (AI) and machine learning (ML) has catalyzed a paradigm shift in biomedical research, offering unprecedented opportunities to decode complex biological systems, accelerate therapeutic discovery, and personalize medical interventions (Guo et al., 2023; Hanna et al., 2025; Yang & Cheng, 2025). AI refers to the simulation of human intelligence by machines, encompassing a broad range of computational techniques, from rule-based expert systems to deep neural networks, capable of learning from data, identifying patterns, and making decisions with minimal human intervention (Aundhia et al., 2024; Cè et al., 2022; Jiménez-Luna et al., 2020). Within the life sciences, ML, a subset of AI focused on algorithmic models that improve through experience, has become instrumental in tackling high-dimensional problems that were previously intractable using classical methods.

6.1. Definition and scope of AI/ML in life sciences

At its core, AI in biomedicine involves the application of computational models to analyze vast datasets generated from genomics, proteomics, metabolomics, imaging, and clinical records (Cè et al., 2022; Esteva et al., 2021; Habib et al., 2025; Jiménez-Luna et al., 2020). These models are trained to recognize patterns, infer relationships, and predict outcomes across diverse biological contexts. In the realm of peptides and proteins, AI encompasses a wide spectrum of methodologies, including supervised learning for classification and regression tasks, unsupervised learning for clustering and anomaly detection, reinforcement learning for iterative optimization, and generative modeling for *de novo* design (Aundhia et al., 2024; Cè et al., 2022; Esteva et al., 2021; Habib et al., 2025; Jiménez-Luna et al., 2020).

The scope of AI extends beyond mere automation; it enables hypothesis generation, mechanistic insight, and decision-making support at every stage of the research pipeline, from early target identification to drug repurposing, biomarker discovery, and clinical outcome prediction. Its integration into structural biology, systems medicine, and synthetic biology has further expanded its applicability, positioning AI as a cornerstone of modern biomedical innovation.

6.2. Applications in drug discovery, protein engineering, and biomarker identification

Deep learning models have transformed virtual screening in drug discovery by accurately predicting essential molecular properties, including solubility, binding affinity, and ADMET profiles (Mamoudou et al., 2025; Pires et al., 2015). Advancements such as AlphaFold2 have significantly progressed the field by providing highly accurate predictions of protein structures, thereby enhancing structure-based drug design with exceptional precision at the atomic level (Bryant et al., 2022; Nussinov et al., 2022; Z. Yang et al., 2023). AI-driven techniques in protein engineering, including sequence-to-function modeling and machine learning-guided directed evolution, have enabled researchers to design proteins with improved stability, specificity, and catalytic activity (Gangwal & Lavecchia, 2024; Saini et al., 2025). These innovations are pertinent for enhancing the pharmacological properties of bioactive peptides, facilitating the development of more effective therapeutic candidates.

Furthermore, AI has demonstrated significant utility in the identification of biomarkers. Integrating complex multi-omics datasets allows machine learning algorithms to identify predictive signatures linked to disease progression, treatment response, or potential adverse events (X. He et al., 2023; Yetgin, 2025). Classifiers developed using longitudinal patient data facilitate the identification of new biomarkers, thereby advancing personalized medicine approaches customized for individual patients.

These applications highlight how AI is not only augmenting but also redefining traditional workflows in biomedical science, offering scalable solutions to long-standing challenges.

6.3. Potential to overcome traditional bottlenecks in peptide discovery

A significant benefit is the capacity to substitute high-throughput screening with virtual alternatives. AI models can evaluate millions of peptide sequences *in silico*, prioritizing those anticipated to demonstrate desired bioactivity prior to any experimental validation (X. He et al., 2023; Ivanenkov et al., 2023; Yetgin, 2025). This markedly decreases the time and expenses linked to physical assays, thereby expediting the initial phases of discovery.

Furthermore, AI improves the forecasting of ADMET properties, which include absorption, distribution, metabolism, excretion, and toxicity (Zhai et al., 2025). Advanced machine learning algorithms, utilizing extensive biochemical and pharmacokinetic datasets, can effectively model these essential characteristics (X. He et al., 2023; Ivanenkov et al., 2023; Yetgin, 2025). AI enhances the probability of success in subsequent development phases by early elimination of peptides exhibiting unfavorable druggability characteristics. *De novo* peptide design represents another transformative capability. Generative models, including variational autoencoders (VAEs), generative adversarial networks (GANs), and transformer-based architectures, facilitate the generation of novel peptide sequences with specific functionalities (Goles et al., 2024; Zhai et al., 2025). These models utilize existing data to generate peptides that optimize desired biological activities while adhering to physicochemical constraints, thereby facilitating innovation.

In addition, artificial intelligence enhances target engagement prediction through the integration of structural information, interaction networks, and functional annotations (Jiménez-Luna et al., 2020; Zhai et al., 2025). This enhances comprehension of peptide interactions with molecular targets, elucidating mechanisms of action and aiding in rational drug design. AI facilitates progress in personalization and precision medicine (Beacher et al., 2021; X. He et al., 2023). Incorporating individual genetic, metabolic, and microbiome profiles allows AI to inform the development of personalized peptide therapeutics designed for specific patient subpopulations. This approach marks a notable advancement in the management of metabolic diseases, as patient-specific factors frequently affect treatment outcomes.

These capabilities establish AI as a transformative element in peptide research, transforming a previously slow and empirical process into a rapid, data-driven, and highly predictive activity.

7. Sources and methodologies in AI-driven peptide discovery

The discovery of bioactive peptides for metabolic diseases, driven by AI, functions through an iterative, multi-stage pipeline, as illustrated conceptually in Fig. 2. This framework begins with *Data Acquisition and Curation*, utilizing various sources such as public databases (e.g., BIOPEP-UWM, ChEMBL), experimental wet-lab results, and literature mining, which together establish the foundational knowledge base. This raw data undergoes *Peptide Representation and Feature Engineering*, converting symbolic sequences into machine-readable formats using techniques that include sequence-based features, predicted structural elements, graph-based representations, and advanced learned embeddings such as ProtBERT and ESM.

These carefully constructed representations function as input for the *AI Model Development and Training* phase, during which various machine learning algorithms, including supervised learning, deep learning architectures such as CNNs, LSTMs, and transformers, as well as generative models like GANs, are trained and optimized. The advanced AI models are subsequently utilized in the *Prediction and Design* phase, facilitating tasks including bioactivity prediction, ADMET profiling, *de novo* peptide design, and peptide-target interaction analysis, resulting in the identification of promising bioactive candidates. These *in silico* predictions undergo rigorous experimental validation and optimization through peptide synthesis, *in vitro* and *in vivo* assays, and structural characterization. This process creates a feedback loop that informs and refines subsequent iterations of AI models and data curation efforts. Successful candidates ultimately advance to *Clinical Translation and Application*, with the objective of developing novel peptide-based therapeutic interventions for metabolic diseases.

7.1. Data acquisition and curation

The foundation of any artificial intelligence (AI)-driven discovery pipeline lies in the quality, diversity, and representativeness of the data used to train and validate predictive models (Hanna et al., 2025; Mullowney et al., 2023; Zhai et al., 2025). In the context of bioactive peptide discovery for metabolic diseases, data acquisition involves collecting structured and unstructured information from a wide range of experimental, computational, and literature-based sources. These datasets serve as the raw material upon which machine learning algorithms learn patterns, infer relationships, and generate novel hypotheses (Hanna et al., 2025; Mullowney et al., 2023; Zhai et al., 2025). However, due to the heterogeneity of biological systems and the complexity of metabolic disorders, effective data curation remains a critical challenge that directly impacts model performance and generalizability.

7.2. Public databases: a treasure trove of annotated peptide data

Numerous publicly accessible databases function as essential resources for bioactive peptide research, providing curated compilations of experimentally validated or computationally predicted sequences, accompanied by varying degrees of annotation and functionality.

BIOPEP-UWM (https://biochemia.uwm.edu.pl/biopep/start_biopep.php) is a comprehensive repository dedicated exclusively to bioactive peptides. This includes comprehensive functional annotations, encompassing activities like antihypertensive, antidiabetic, antioxidant, and immunomodulatory effects (Minkiewicz et al., 2019, 2022). BIOPEP-UWM offers tools for analyzing peptide profiles and predicting potential bioactivities based on sequence similarity, thus serving as a valuable resource for discovery and characterization.

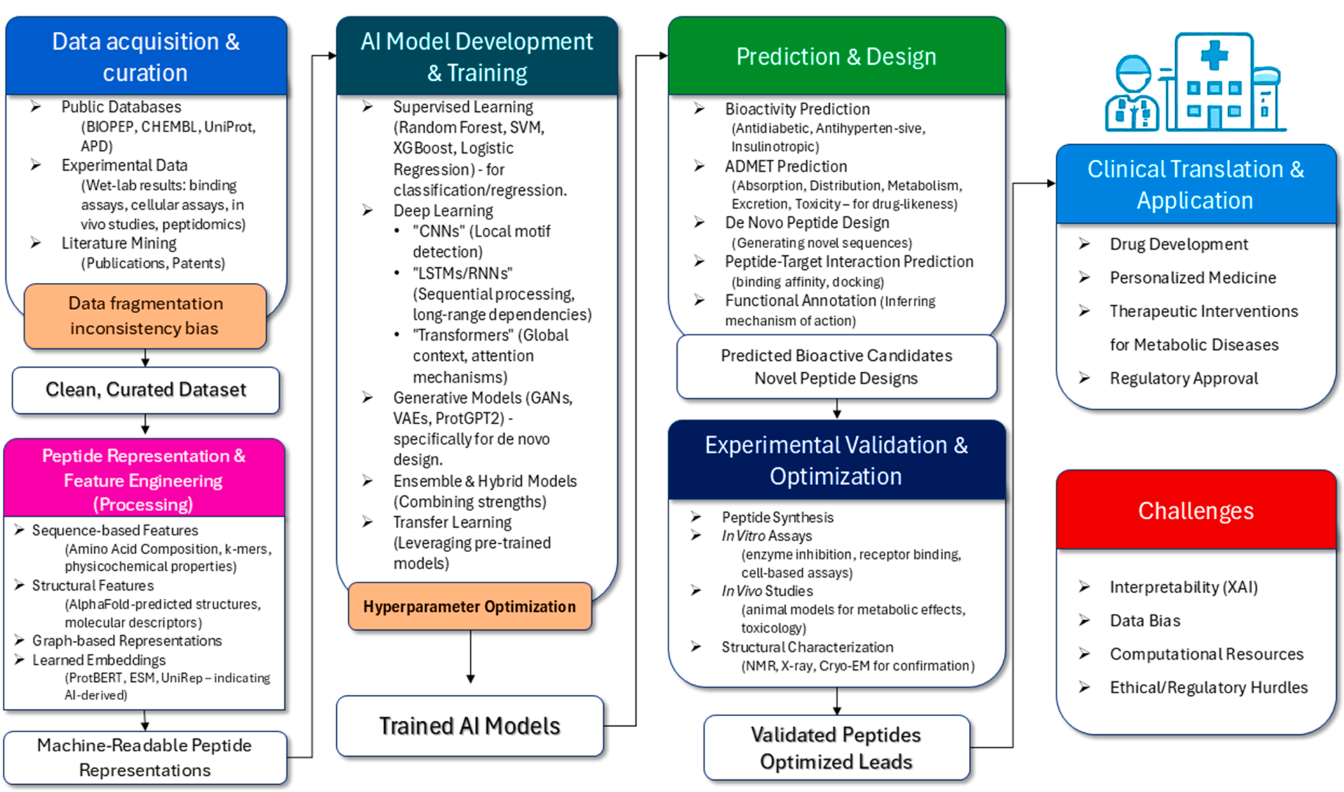

Fig. 2. Conceptual framework of the AI-driven bioactive peptide discovery pipeline for metabolic diseases.

The Antimicrobial Peptide Database (APD) (https://aps.unmc.edu) primarily focuses on antimicrobial peptides and provides extensive structural and physicochemical data. This information is essential for comprehending peptide stability, charge distribution, and membrane interaction (attributes that are also significant for metabolic peptides) (Bin Hafeez et al., 2021). Consequently, APD functions as a valuable resource for the design of peptides with enhanced pharmacological properties.

PeptideRanker (http://distilldeep.ucd.ie/PeptideRanker/) is a significant resource that utilizes a machine learning-based scoring system to assess the probability of a peptide exhibiting bioactivity (Pearman et al., 2020). PeptideRanker, trained on a variety of peptide sequences with defined functions, is particularly useful for the initial screening of candidate peptides, allowing researchers to prioritize sequences that exhibit a greater likelihood of bioactivity (Pearman et al., 2020).

ChEMBL (https://www.ebi.ac.uk/chembl/) offers a comprehensive database of bioactive molecules, including small peptides, along with relevant pharmacological and ADMET data, facilitating extensive drug discovery initiatives (Davies et al., 2015; Zdrazil et al., 2024). ChEMBL plays a crucial role in identifying peptides that have established target interactions and clinical significance, thereby connecting peptide research with therapeutic development (Davies et al., 2015; Zdrazil et al., 2024).

Moreover, UniProt (https://www.uniprot.org/) serves as a central repository for protein sequence and functional data, enabling researchers to obtain endogenous peptides from precursor proteins that play a role in metabolic regulation (UniProt Consortium, 2023).

These databases are essential for training supervised machine learning models and evaluating new algorithms in peptide research. Nonetheless, they possess certain limitations. A significant number of individuals exhibit biases towards extensively studied peptides, face challenges due to incomplete functional annotations, and encounter underrepresentation of peptides that target metabolic pathways. Addressing these gaps is essential for advancing the field and ensuring comprehensive and equitable coverage of bioactive peptides in future research.

The proliferation of public databases containing peptide-related information has facilitated advancements in AI-driven discovery. However, the efficacy of these databases in supporting AI applications varies substantially, contingent upon their content, annotation granularity, and domain specificity. A rigorous assessment of these resources is crucial for researchers to judiciously select datasets for training and validating predictive models, as this directly influences model generalizability and accuracy.

Table 1 provides a comparative overview that emphasizes the fragmented but complementary nature of publicly available databases and computational resources that support AI-driven bioactive peptide discovery. This heterogeneity presents both an opportunity and a considerable obstacle in the advancement of next-generation biotherapeutics aimed at metabolic diseases from a translational perspective. Peptide-centric repositories like BIOPEP-UWM, AHTPDB, and DPPIVPdb offer a substantial collection of experimentally validated leads that are directly pertinent to enzymatic targets crucial for cardiometabolic health, including ACE, DPP-IV, α-amylase, and α-glucosidase (Minkiewicz et al., 2019, 2022). These resources exhibit inherent biases favoring food-derived peptides and well-characterized targets, thereby constraining the exploration of chemical space and impeding the identification of non-canonical modulators with novel mechanisms of action.

In contrast, comprehensive repositories like ChEMBL and UniProt offer detailed pharmacological and mechanistic annotations, facilitating the incorporation of peptides into systems-level biological networks (Davies et al., 2015; UniProt Consortium, 2023; Zdrazil et al., 2024). These resources are not inherently optimized for peptide therapeutics and typically necessitate advanced computational processing (precursor–fragment mapping, evidence grading, leakage-free clustering) prior to the implementation of effective downstream AI modeling. Antimicrobial peptide-focused databases (APD, DRAMP, DBAASP) provide meticulously curated physicochemical and safety annotations that are essential for constructing transferable representations (stability, membrane interactions, hemolysis) (Bae et al., 2025; Bin Hafeez et al., 2021). However, their activity profiles frequently differ from metabolic therapeutic targets.

The findings highlight the need for data harmonization pipelines that can integrate diverse labels, standardize assay metadata, and systematically resolve data redundancy and class imbalance. To ensure that AI frameworks effectively prioritize bioactive peptides with therapeutic potential, it is essential that they are trained on datasets that include not only positive enzymatic activity labels but also contextual features such as digestion stability, protease susceptibility, and off-target safety. The integration of task-specific resources, such as AHTPDB and DPPIVPdb, with extensive knowledge bases like ChEMBL and UniProt in multi-objective learning architectures provides a systematic approach to effective candidate prioritization for metabolic disease indications.

The commentary derived from Table 1 indicates that a cohesive, evidence-based data infrastructure is essential for bridging the existing translational gap in next-generation biotherapeutics. Embedding multi-source knowledge into deep learning frameworks, spanning self-supervised peptide language models to multi-task predictive networks, facilitates a transition in AI-driven discovery from narrow enzyme inhibition screening to the rational design of multifunctional peptides that can modulate complex metabolic networks. This aligns with the main thesis of the article: the convergence of curated data, advanced AI, and integrative bioinformatics will redefine the future landscape of bioactive peptide therapeutics for metabolic diseases.

7.3. Experimental datasets from proteomics, peptidomics, and metabolomics studies

In addition to curated databases, high-throughput experimental datasets produced by contemporary "-omics" technologies offer a substantial source of empirical data that illustrates the dynamic and context-dependent characteristics of peptide biology.

Proteomics studies are essential for identifying and quantifying full-length proteins expressed in specific physiological or pathological conditions (Shah & Misra, 2011). The datasets provide insights into precursor proteins that can generate bioactive peptides via proteolytic cleavage, elucidating the origins and regulatory mechanisms of peptide production (Husi & Albalat, 2014; Kaczor-Urbanowicz & Wong, 2020; Shah & Misra, 2011).

Peptidomics, a specialized field, concentrates on the profiling of endogenous peptides found in tissues, biological fluids, or cellular compartments (Hellinger et al., 2023). Mass spectrometry (MS)-based peptidomics has proven essential in identifying novel signaling peptides involved in key processes including glucose metabolism, appetite regulation, and inflammation (Hellinger et al., 2023). This method facilitates the identification of peptides that may have therapeutic significance in metabolic disorders.

Metabolomics enhances these initiatives by analyzing small molecule metabolites affected by peptide activity. Through the analysis of downstream effects, metabolomics offers a comprehensive understanding of the ways in which peptides influence metabolic networks and engage with wider biological pathways (Avuthu et al., 2024; He et al., 2023; Yetgin, 2025). These multi-omics approaches collectively provide a detailed understanding of peptide biology, connecting molecular mechanisms to functional outcomes.

The integration of multi-layered datasets enables AI models to move beyond static sequence analysis and include functional context, such as tissue-specific expression patterns, post-translational modifications, and disease-related perturbations. Nonetheless, realizing this integration presents considerable difficulties. Advanced data preprocessing pipelines are necessary to normalize, annotate, and align heterogeneous data types across various platforms and studies. Overcoming these technical

Table 1
Comparative overview of key public resources for bioactive peptide discovery.

Database	Primary content & focus	Key metadata useful for AI	Strengths for AI/ML applications	Typical pitfalls / limitations
(A) Peptide-centric databases and tools				
BIOPEP-UWM	Experimentally validated food-derived bioactive peptides; *in silico* tools (e.g., bioactivity prediction, enzyme cutting patterns).	Sequence; reported activities (ACE, DPP-IV, antioxidant, etc.); source protein; digestion patterns.	Rich starting point for metabolic peptide leads; digestion/enzyme modules help generate training negatives/decoys; supports sequence-activity mapping.	Bias toward food proteins and well-studied activities; mixed evidence quality; redundancy across variants/fragments.
APD (Antimicrobial Peptide Database)	Antimicrobial peptides with structures, physicochemical properties, activities, and some targets/pathogens.	Charge, hydrophobicity, length, modifications; curated activity labels; sometimes 3D or structural class.	High-quality property labels for learning sequence → property mappings (charge, amphipathicity, helicity) transferable to permeability/ stability models.	Activity space is mainly antimicrobial; limited relevance to metabolic enzymes; class imbalance (many actives, scarce labeled inactives).
DRAMP (Database of Research on Antimicrobial Peptides)	Extensive AMP entries incl. natural/synthetic, hemolysis, toxicity notes, and some structural data.	Hemolysis/toxicity flags; MICs; sequence variants.	Valuable safety signals (hemolysis) for multi-objective design; broad coverage of modifications.	Similar domain bias as APD; heterogeneous assay protocols.
DBAASP (Database of Antimicrobial Activity and Structure of Peptides)	Quantitative activity data against organisms/membranes; structural classes.	MICs (quant.); salt, pH, medium; secondary structure annotations.	Offers quantitative labels for regression and uncertainty calibration; good for learning assay-context effects.	Predominantly antimicrobial; variability in assay conditions.
SATPdb (Structurally Annotated Therapeutic Peptides)	Therapeutic peptides with 3D structural annotations and modifications.	3D info (PDB links/models), modifications, targets.	Facilitates structure-aware featurization (distance maps, torsions) and docking constraints; exposure to PTMs.	Coverage uneven across indications; quality varies for modeled structures.
PeptideRanker	ML-based score predicting “general bioactivity” from sequence.	Probability score; sequence input only.	Fast pre-filter for large peptide libraries prior to task-specific models.	Task-agnostic; score is not mechanism-specific; prone to false positives; not a replacement for domain models.
Hemolytik / HemoPI (hemolysis)	Hemolytic/non-hemolytic peptide data.	Binary/quant. hemolysis labels; sequences.	Useful to guardrail cytotoxicity in multi-objective optimization.	Assay heterogeneity; may not reflect *in vivo* tolerability.
ToxinPred / ToxinPred2 (toxicity prediction)	*In silico* toxicity classifiers for peptides.	Predicted toxicity scores.	Triage tool to filter toxic motifs before synthesis.	Model bias; predictions depend on training set composition; confirm experimentally.
(B) Narrow domain & clinical peptide resources (high relevance to metabolic and cardiovascular axes)				
Database	**Primary content & focus**	**Key metadata**	**Strengths for AI/ML**	**Pitfalls**
AHTPDB (Antihypertensive Peptide DB)	ACE-inhibitory peptides and related antihypertensive activities.	IC_{50}s/activities vs ACE; source proteins; sequences.	Direct labels for ACE inhibition, ideal for training/ranking ACE-targeted candidates.	Often small, skewed datasets; limited diversity beyond ACE.
DPPIVPdb / DPP-IV peptide DB	Dipeptidyl peptidase-IV inhibitory peptides.	Activity/IC_{50} vs DPP-IV; sequences; food sources.	Mechanism-specific labels for antidiabetic pipelines.	Dataset size modest; assay contexts vary.
THPdb	FDA-approved & clinical trial peptides (sequences, indications, delivery).	Indication, formulation, route, clinical status.	Ground truth for developability features (length, PTMs, delivery); helps define realistic design space.	Focus on advanced peptides; not comprehensive for discovery-stage actives.
CAMP / CAMPR3	Anti-microbial peptide resource with prediction tools.	Activity classes; physicochemical descriptors.	Additional descriptor baselines and activity labels.	Similar AMP bias; heterogeneous labels.
(C) General knowledge bases, targets, pathways, ADMET, provenance				
Resource	**Primary content & focus**	**Key metadata**	**Strengths for AI/ML**	**Pitfalls**
ChEMBL	Bioactivity data (ligand–target assays) including peptide ligands; ADMET fields; clinical phases.	Assay types, targets (UniProt), pChEMBL, mechanisms, off-targets.	Links peptide sequences/analogues to curated targets and assay outcomes; excellent for off-target mining and label harmonization.	Needs careful filtering for peptide classes; peptide–small-molecule mixing can confound featurization.
BindingDB	Quantitative binding data for ligand–protein interactions (some peptides).	Kd/Ki/IC_{50}; target species; conditions.	Adds quantitative anchors for enzyme inhibition (ACE, DPP-IV) when available.	Historically small peptide coverage; assay variability.
UniProt (Swiss-Prot/ TrEMBL)	Canonical protein sequences, annotations, PTMs, function, taxonomy.	Protein/precursor IDs; features (signal peptides, domains); PTMs.	Essential to trace peptide provenance (precursor proteins, PTMs), map to pathways and organism sources.	Not a peptide DB; requires computational extraction of bioactive fragments; extensive preprocessing.
MEROPS (peptidases)	Proteases/peptidases and their substrates.	Enzyme families; cleavage rules; substrate motifs.	Enables *in silico* digestion and stability modeling (protease susceptibility) for oral/serum contexts.	Translation from motif to *in vivo* stability requires care.
BRENDA / Reactome / KEGG	Enzymes & pathways (metabolic context).	Enzyme kinetics; pathway membership.	Connects peptide targets to metabolic networks (e.g., DPP-IV in incretin axis).	Licensing/use constraints vary; harmonization needed.
PRIDE / PeptideAtlas	Mass-spec identifications of peptides/proteins.	PSMs, PTMs, sample context.	Real-world detectability constraints; helps negative sampling (peptides not observed).	Not activity-labeled; requires downstream linkage.

How to read this table 1: I group resources into (A) peptide-centric databases, (B) narrow domain/clinical peptide resources, and (C) general knowledge bases that help map peptides to targets, pathways, and ADMET risks. “AI/ML strengths” highlights what each source contributes to data-driven pipelines; “Pitfalls” flags issues you should mitigate during curation. Where license/API specifics vary by release, I note them qualitatively (avoid relying on exact legal text in code).

challenges is crucial for realizing the complete potential of AI-driven peptide discovery and facilitating a more profound comprehension of peptide biology in health and disease.

7.4. Integration of multi-omics data for enhanced predictive power

One of the most promising advancements in AI-driven peptide discovery is the integration of multi-omics data, which combines genomics, transcriptomics, proteomics, peptidomics, and metabolomics into a unified analytical framework. This consolidative approach allows for the development of context-aware models that not only capture sequence-function relationships but also account for regulatory networks and environmental influences. For instance, transcriptomic data can uncover gene expression profiles in metabolic tissues such as the liver, adipose tissue, and pancreas under various disease states (Erfanian et al., 2023; Gomase & Tagore, 2008). Meanwhile, proteomic and peptidomic data enable the mapping of active peptides to their parent proteins and cellular locations, providing insights into their functional roles (Hellinger et al., 2023; Kaczor-Urbanowicz & Wong, 2020; T.R. Shah & Misra, 2011). Additionally, metabolomic analyses link peptide activity to downstream metabolic outcomes, including glucose uptake, fatty acid oxidation, and inflammatory cytokine production (Avuthu et al., 2024). By integrating these complementary layers of information, AI models can be trained to predict not only whether a peptide is bioactive but also the specific conditions under which it exerts its effects, thereby enhancing both translational relevance and mechanistic understanding.

Despite its potential, integrating multi-omics data poses significant challenges. Data harmonization is complicated by the fact that different omics platforms generate data at varying resolutions, formats, and scales. Furthermore, experimental variability and technical limitations often result in missing or noisy data points, while the high dimensionality and interdependence of these datasets demand scalable algorithms and robust statistical frameworks.

To address these issues, researchers are increasingly turning to advanced computational techniques, such as data imputation, dimensionality reduction, and deep learning architectures capable of handling multimodal inputs (autoencoders, graph neural networks (GNNs), and attention-based transformers).

8. Feature engineering in peptide representation

Feature engineering plays a pivotal role in the success of artificial intelligence (AI) models applied to bioactive peptide discovery. Since peptides are fundamentally sequences of amino acids, transforming these symbolic sequences into meaningful numerical representations that capture relevant biological and physicochemical properties is essential for effective machine learning. The choice of feature representation not only influences model performance but also determines how well the AI system can generalize across unseen data and interpret underlying biochemical principles.

8.1. Sequence-based features: capturing local and global patterns

Sequence-based feature engineering is essential for early-stage peptide AI, employing three complementary strategies (Fig. 3). Amino Acid Composition (AAC) offers fundamental residue frequency metrics that balance interpretability with limitations related to compositional bias; k-mer frequency analysis identifies local functional motifs through di-/tripeptide patterns to forecast bioactive sites and molecular interactions (Ali et al., 2024; Moeckel et al., 2024); and the integration of physicochemical properties quantifies structural-electronic attributes (hydrophobicity, charge dynamics, conformational flexibility) using database-informed averaging (AAindex, ProtParam (https://web.expasy.org/protparam/)) to model bioavailability and target specificity (Schwede et al., 2003).

These features are typically used in traditional machine learning models such as Support Vector Machines (SVM), Random Forests, and Gradient Boosting Trees, which rely on handcrafted inputs rather than

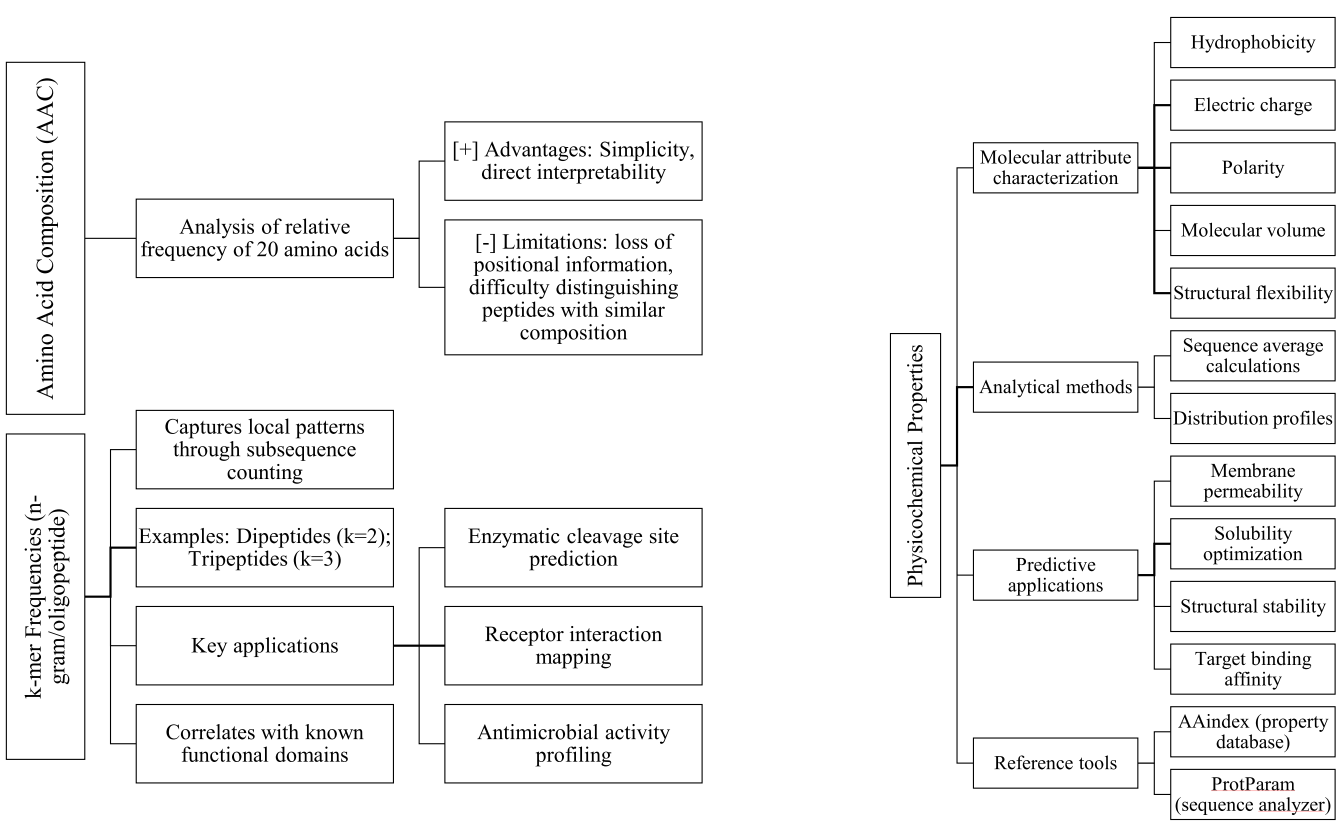

Fig. 3. Sequence-based features in peptide AI models.

end-to-end learning (Qiu et al., 2020; Sanders et al., 2011; Tripathi et al., 2021).

This subsection introduces sequence-based features as fundamental for early-stage peptide AI. The discussion emphasizes the utility of Amino Acid Composition (AAC) and k-mer analysis for interpretability and the identification of local motifs. However, it critically notes their main limitation: reliance on handcrafted inputs, which constrains their capacity to capture complex, non-linear relationships and long-range dependencies that are fundamental to peptide bioactivity. This inherently restricts their application mainly to conventional machine learning models and frequently requires substantial domain expertise for effective feature engineering.

8.2. Structural features: incorporating conformational context

The peptide sequence offers essential insights; however, its biological function is largely determined by its three-dimensional conformation, especially in interactions with protein targets. Computational methods are utilized to predict structural features that affect functionality within this spatial context.

Secondary structure prediction tools evaluate the likelihood that peptide segments assume α-helical, β-sheet, or coil conformations (Pirovano & Heringa, 2010; Singh et al., 2019). Specific structural motifs, such as amphipathic helices, are closely linked to particular functions, including cell penetration and antimicrobial activity. Hydrophobicity profiles along the peptide backbone assist in identifying regions that participate in membrane interaction or receptor binding (Pirovano & Heringa, 2010; Singh et al., 2019).

Furthermore, predictions regarding flexibility and solvent accessibility, utilizing algorithms such as B-factor or DSSP-derived measures, offer insights into the accessibility of specific residues for enzymatic cleavage or ligand interaction (X. Xu & Zou, 2022). These features are especially pertinent for modeling dynamic processes, including receptor-ligand binding and enzyme inhibition.

The analysis of structural features highlights the significant importance of three-dimensional conformation in peptide function, thereby improving predictive abilities for complex biological interactions. The critical issue lies in the trade-off, although these features offer extensive context, they require substantial computational resources for precise predictions. Additionally, inaccuracies in structural modeling may introduce noise, which can undermine the robustness and reliability of subsequent AI models. This emphasizes a continual challenge in reconciling data richness with computational feasibility and the quality of predictions.

The integration of structural features markedly improves the predictive capabilities of AI models, particularly in contexts involving intricate biological interactions, including receptor-ligand binding, enzyme inhibition, and blood-brain barrier penetration. These predictions require significant computational resources and may generate noise if the accuracy of the structural predictions is inadequate. Therefore, although structural features provide important context, their integration requires careful balancing to maintain robust and reliable model performance.

8.3. Graph-based representations: modeling residue interactions

A novel approach in peptide representation is the modeling of peptides as graphs, with nodes representing amino acid residues and edges denoting physical or inferred interactions. This framework, based on graph theory, enables the application of advanced machine learning techniques, including graph neural networks (GNNs) and various topological learning methods, to effectively capture intricate structural and functional relationships (Imre et al., 2025; Lei et al., 2024; Srivastava et al., 2024).

Molecular graphs represent each residue as a node characterized by attributes such as type, charge, and hydrophobicity, with edges indicating covalent or non-covalent interactions. Graph Neural Networks (GNNs) can extract hierarchical patterns from graphs, effectively capturing long-range dependencies and spatial relationships that traditional sequence-based approaches often overlook (Imre et al., 2025; Srivastava et al., 2024). Residue interaction networks, obtained from molecular dynamics simulations or predicted contact maps, highlight residue-residue interactions that are essential for elucidating structure-function relationships (Imre et al., 2025; Srivastava et al., 2024). These networks are essential for examining allosteric effects and pinpointing critical residues for mutational analysis.

The introduction of graph-based representations serves as a significant analytical advancement, acknowledging their superior ability to model intricate residue interactions and spatial relationships that simpler sequence-based methods frequently neglect. The critical analysis accurately identifies the challenges involved; these advanced models generally require larger datasets for effective training and significantly increased computational resources. Despite these demands, their ability to integrate comprehensive structural and interaction data establishes them as a significant, though resource-intensive, pathway for advanced peptide research.

Graph-based models offer an enhanced understanding of peptide behavior by connecting sequence information with three-dimensional structure (Imre et al., 2025; Lei et al., 2024; Srivastava et al., 2024). Nonetheless, their implementation presents challenges. In contrast to simpler feature sets, these models necessitate larger datasets for robust performance and require substantially greater computational resources. Despite these limitations, their capacity to integrate comprehensive structural and interaction data renders them an effective instrument for advancing peptide research.

8.4. Embeddings: learning distributed representations

The advent of deep learning has catalyzed the adoption of embedding-based representations, which excel at capturing complex, high-level abstractions directly from raw peptide sequences without the need for explicit feature engineering (Fig. 3). This method has transformed the processing and analysis of peptide data. Word2Vec and Skip-Gram models, derived from natural language processing (NLP), conceptualize amino acids as "words" within a sequence, learning dense vector representations informed by their contextual usage across extensive peptide datasets (V. Kumar et al., 2025; Sharma et al., 2021; C. Wu et al., 2019). These embeddings represent semantic similarity, with structurally or functionally related residues clustering in the embedding space. This facilitates the identification of biologically significant patterns autonomously.

ProtBERT and ESM (Evolutionary Scale Modeling) are advanced transformer-based language models that have been pre-trained on comprehensive protein sequence databases, including UniRef and Pfam (N. Kumar et al., 2025; F. Zhang et al., 2024). These models produce contextualized embeddings that capture evolutionary, structural, and functional signals. ProtBERT, which utilizes the BERT architecture, and ESM have shown remarkable transferability across various downstream tasks, such as secondary structure prediction, functional annotation, and bioactivity classification (N. Kumar et al., 2025; F. Zhang et al., 2024).

Recently, transformer-based language models have advanced the field by utilizing complete transformer architectures to analyze entire peptide sequences (Lai et al., 2025; Ruan et al., 2025; F. Zhang et al., 2024). These models provide attention-based interpretations, effectively capturing long-range dependencies, identifying conserved motifs, and generating novel sequences with specified properties. The capacity to model intricate relationships within sequences renders them especially effective for peptide discovery and design.

Embeddings allow deep learning models to gain high-level abstractions directly from raw sequences, eliminating the need for explicit feature engineering. Pre-trained models (ProtBERT, ESM) capture contextual information well and generalize well across tasks, but their "black-box" nature is a drawback. For biomedical applications, explainable AI (XAI) tools are needed to interpret learnt representations and integrate predictive skills with mechanistic knowledge.

Embedding-based methods remove the need for manual feature selection, enabling models to derive abstract representations directly from data. Their strength is found in their generalizability across tasks and their ability to utilize extensive unlabeled sequence databases via self-supervised learning. Pre-training on extensive protein sequence corpora enables these models to derive significant biological insights, which can be further refined for targeted applications, establishing them as fundamental to contemporary peptide research.

9. Machine learning (ML) algorithms applied

The application of machine learning (ML) algorithms has become central to the discovery and characterization of bioactive peptides, particularly in the context of metabolic disease prevention (Alqarni & Alqarni, 2025; Cè et al., 2022; Kantor & Morzy, 2024; Y. Yang & Cheng, 2025; Yetgin, 2025; Zhao et al., 2022). These algorithms leverage diverse peptide representations, ranging from simple sequence features to complex embeddings, to predict functional properties, classify bioactivity types, and optimize therapeutic candidates. The choice of algorithm depends on data characteristics, problem complexity, interpretability requirements, and computational resources.

9.1. Supervised learning: foundational models for classification and regression

Supervised learning remains essential in the prediction of early-stage bioactive peptides due to its simplicity, interpretability, and efficacy with smaller, well-annotated datasets. These models utilize labeled examples, such as active and inactive peptides, to generalize learned patterns for predicting the properties of novel sequences. One commonly employed method is Random Forest (RF), an ensemble technique that integrates multiple decision trees to enhance resilience against overfitting (Bizzotto et al., 2024; Tripathi et al., 2021). RF demonstrates strong performance in high-dimensional feature spaces and offers feature importance rankings, thereby serving as an effective method for pinpointing essential amino acid positions or physicochemical properties that influence peptide activity (Bizzotto et al., 2024; Tripathi et al., 2021). Support Vector Machines (SVM) have demonstrated efficacy in high-dimensional data contexts. Utilizing kernel tricks, SVMs effectively identify non-linear relationships between features and outcomes, facilitating classification tasks like differentiating antidiabetic peptides from inactive counterparts (Sanders et al., 2011).

XGBoost is a gradient boosting algorithm recognized for its efficiency and effectiveness, especially in managing sparse and imbalanced datasets (Cai et al., 2017; Charoenkwan et al., 2022; Zhang et al., 2024). XGBoost has been effectively utilized to predict DPP-IV inhibitory peptides and other enzyme-targeting molecules associated with glucose regulation (Zhang et al., 2024). Logistic regression serves as a widely utilized baseline model, characterized by its transparency and interpretability (Jabbar et al., 2020). Logistic regression, when utilized alongside regularization techniques such as LASSO, can effectively identify minimal predictive subsets of features, thereby elucidating the most significant factors influencing peptide activity.

Supervised learning models perform well in structured environments; however, they frequently depend on substantial feature engineering to preprocess raw sequence data. They also encounter difficulties in capturing long-range dependencies present in peptide sequences, which may restrict their capacity to fully utilize the complexity of biological data. Despite these limitations, their interpretability and performance render them essential tools in numerous peptide discovery workflows.

9.2. Deep learning: end-to-end modeling of peptide function

Deep learning architectures have transformed peptide modeling by enabling end-to-end learning, where raw sequences are directly mapped to functional predictions without the need for manual feature extraction (Cao et al., 2020; Erfanian et al., 2023; Thieme et al., 2023). These models are particularly adept at uncovering complex, hierarchical patterns within large-scale datasets, making them highly effective for peptide discovery and analysis.

For instance, Convolutional Neural Networks (CNNs), originally designed for image analysis, have been adapted to detect local motifs within peptide sequences using filters that slide across the sequence (Thieme et al., 2023; S. Zhang & Li, 2022). This capability makes CNNs especially effective for identifying short functional signatures, such as receptor-binding domains or enzymatic cleavage sites, which are critical for understanding peptide activity.

In contrast, Recurrent Neural Networks (RNNs) and their advanced variant, Long Short-Term Memory (LSTM) networks (Y. Li et al., 2022; Sharma et al., 2021), process peptides as sequential inputs, maintaining a memory of previous residues to model long-range dependencies. LSTMs have demonstrated significant potential in predicting peptide ADMET properties and classifying multi-functional activities, where context and sequence order play a crucial role.

More recently, transformer-based models, inspired by Natural Language Processing (NLP), have emerged as a powerful tool in peptide research (Kumar et al., 2025). These models employ self-attention mechanisms to dynamically weigh the importance of different residues across the entire sequence, capturing both local and global context simultaneously. Transformers pre-trained on massive protein databases (ProtBERT, ESM, and UniRep) generate contextualized embeddings that encode structural and functional information (Zhang et al., 2024). These embeddings significantly enhance performance in downstream tasks, including bioactivity prediction, secondary structure modeling, and function annotation.

In addition to predictive tasks, deep learning has transformed *de novo* peptide design via generative models, particularly Generative Adversarial Networks (GANs). GANs function based on a min-max game principle, comprising a generator network that produces synthetic peptide sequences and a discriminator network that learns to differentiate these synthetic outputs from actual peptide data (Goles et al., 2024; Lai et al., 2025; Lin et al., 2022). Through iterative competition, the generator develops proficiency in producing novel sequences that closely resemble the distribution of real bioactive peptides, effectively acquiring the structural patterns of functional peptides. This distinctive adversarial training allows GANs to navigate extensive chemical and sequence spaces, producing peptides with particular desired characteristics (e.g., binding affinity, stability, or target specificity) without the need for explicit, manually crafted rules (Goles et al., 2024; Lai et al., 2025; Lin et al., 2022). The capacity of GANs enables them to serve as effective instruments for investigating novel chemical spaces, facilitating the identification of new therapeutic candidates that may elude conventional screening or optimization techniques.

Deep learning models offer superior scalability and generalization when trained on sufficiently large and diverse datasets, making them ideal for handling the complexity of peptide biology. However, they also come with challenges, such as high computational resource demands and reduced interpretability compared to traditional methods. Despite these limitations, their ability to learn intricate patterns directly from data has positioned deep learning as a cornerstone of modern peptide research.

9.3. Ensemble and hybrid models: combining strengths for improved accuracy

In order to address the drawbacks of individual models, it is possible to employ hybrid and ensemble approaches. These methods combine several algorithms or combine deep learning components with rule-based systems. These tactics concentrate on correcting the shortcomings of many approaches while utilizing the finest of them. One method is stacking, which involves training many base learners, such as Convolutional Neural Networks, Random Forests, or Support Vector Machines, and then combining their output using a meta-classifier (Sanders et al., 2011; Tripathi et al., 2021; Zhang & Li, 2022). By combining the advantages of each model, this method improves forecast accuracy and dependability.

To gain a deeper understanding of the structure and function of peptides, another method combines many model types into hybrid structures. CNNs and LSTM networks can collaborate to identify local patterns while comprehending the context of the entire sequence (Y. Li et al., 2022). When combined with attention processes, Graph Neural Networks (GNNs) can enhance our comprehension of long-range relationships and residue interactions (Imre et al., 2025; Srivastava et al., 2024). By examining both the local and global characteristics of peptides, these designs aid in our understanding of them.

Rule-based and machine learning hybrids use domain knowledge into their machine learning procedures, such as enzyme cleavage rules or well-known pharmacophores (Jiang et al., 2022; Zhang et al., 2024). These models are particularly helpful when data is few or comprehension of the results is crucial because of this integration, which increases biological relevance and reduces false positives.

In crucial domains such as clinical candidate screening, ensemble and hybrid approaches are gaining popularity and demonstrating excellent outcomes in benchmark studies. To properly handle the intricacies of peptide biology, they employ a variety of techniques, improving prediction accuracy and dependability.

9.4. Transfer learning approaches using pre-trained language models

A transformational innovation in AI-driven peptide discovery is transfer learning, which includes fine-tuning models pre-trained on massive unlabeled protein sequence datasets for specific purposes using smaller, labeled datasets. This method provides a quick means to use already available data and reflects how people use past knowledge to tackle fresh challenges. Among the notable examples are ProtBERT, a BERT-based model trained on millions of protein sequences creating contextualized embeddings encapsulating functional and evolutionary information (Yetgin, 2025; Zhang et al., 2024). Likewise, ESM (Evolutionary Scale Modeling) use masked language modeling to build robust representations and has demonstrated remarkable accuracy in protein structure and function prediction. UniRep is another basic LSTM-based model that effortlessly combines into several downstream prediction tasks and captures sequence-to- function mappings (Li et al., 2022; Sharma et al., 2021). In few-shot learning, these pre-trained models shine and provide excellent forecasts even with few experimental data.

Table 2
Comparative analysis of machine learning and deep learning algorithms in AI-driven bioactive peptide discovery.

Algorithm type	Typical peptide discovery tasks	Strengths	Limitations	Usability & implementation considerations
Supervised Learning (Random Forest, Support Vector Machines, XGBoost, Logistic Regression)	• Bioactivity classification/ regression (e.g., ACE/DPP-IV inhibition) • ADMET prediction (solubility, hemolysis, protease stability) • Peptide–target interaction scoring • Prioritization for *in vitro* screening	• Interpretability: Transparent feature importance enables mechanistic insights • Robustness: Performs well on small-to-moderate, structured datasets • Computationally efficient: Low training cost; suitable for rapid iteration	• Feature engineering–dependent: Requires handcrafted descriptors (e.g., amino acid composition, physicochemical indices) • Limited scalability: Struggles with raw sequence data and long-range dependencies • May underperform in novel pattern discovery beyond engineered features	• Highly accessible (Scikit-learn, XGBoost) • Ideal for initial screening and hypothesis testing • Effective when combined with biologically meaningful descriptors (e.g., cleavage motifs, binding pocket scores)
Deep Learning (CNNs, RNNs/LSTMs, Transformers)	• *De novo* peptide design (generative models, VAEs, GANs) • Multi-functional peptide prediction (e.g., dual ACE/ DPP-IV inhibitors) • Complex sequence–activity mapping • Automated extraction of structural and evolutionary features	• Automated feature extraction: Captures hierarchical sequence/ structure patterns directly from raw data • Scalability & accuracy: State-of-the-art performance on large, heterogeneous datasets • Global context capture: Transformers excel at modeling long-range dependencies in peptide sequences	• Reduced interpretability: Often considered "black-box" models • Data hungry: Requires large, high-quality labeled datasets • High computational cost: GPU-intensive; overfitting risk on small or biased data • Requires extensive hyperparameter optimization	• Implemented via TensorFlow, PyTorch; growing ecosystem for bioinformatics (e.g., DeepChem) • Best suited for complex design tasks and large datasets • Explainability techniques (e.g., SHAP, attention maps) can partially address interpretability gaps
Ensemble and Hybrid Models (stacked classifiers, CNN–LSTM hybrids, multi-head Transformers)	• Enhanced bioactivity prediction (classification & regression) • Multi-objective optimization (e.g., potency + stability + safety) • ADMET property modelling • Reduction of variance across models	• Improved accuracy & robustness by combining complementary algorithms • Mitigates individual model weaknesses • Flexibility to integrate domain knowledge (e.g., known motifs, cleavage patterns) • Can balance generalization and interpretability	• Higher complexity: Challenging to optimize and maintain • Computationally more demanding (training multiple models) • Interpretability varies depending on ensemble design	• Often custom-built; requires careful architecture selection (bagging, boosting, stacking) • Effective for maximizing predictive power in high-stakes tasks (e.g., lead prioritization before costly synthesis)
Transfer Learning (pre-trained models such as ProtBERT, ESM, UniRep, AlphaPept)	• Bioactivity prediction with limited labeled data • Function annotation (binding target, enzymatic pathway) • Few-shot learning for rare peptide classes (e.g., cyclic peptides, PTM-rich peptides) • Embedding generation for downstream tasks	• Leverages large-scale unlabeled protein data: Provides rich sequence embeddings capturing evolutionary/structural context • High performance on small datasets through fine-tuning • Produces generalizable representations transferable across targets	• Pre-training is computationally prohibitive, though fine-tuning is feasible • Domain shift risk: Performance drops if fine-tuning data differs greatly from pre-training data (e.g., metabolic vs. antimicrobial peptides) • Models can be very large, slowing inference in production	• Fine-tuning supported by Hugging Face Transformers, PyTorch Lightning • Highly recommended when labeled data are sparse • Combine with active learning or task-specific layers to reduce domain shift

Their capacity to extend across species, purposes, and modalities makes them priceless for hastening the synthesis of new peptides. In contemporary peptide research, they are becoming indispensable instruments by bridging gaps in data availability and biological knowledge.

The efficacy of artificial intelligence in bioactive peptide discovery is contingent upon the judicious selection and rigorous evaluation of machine learning and deep learning algorithms. The suitability of each algorithmic paradigm is dictated by the specific requirements of the peptide discovery task, including the type of prediction or design task (e.g., sequence-based prediction, *de novo* design, or ADMET profiling), the quantity and complexity of available data, and the desired level of model interpretability and transparency. A comparative analysis of these methodologies is presented in Table 2, which delineates their typical applications, limitations (, limitations (including data requirements, computational complexity, and opacity), and practical considerations for their effective implementation in a research context. This comparative framework is essential for researchers seeking to optimize model performance, address specific biological questions, and navigate the inherent trade-offs associated with the development of complex AI models.

Table 2 highlights the necessity for algorithm selection to be strategically aligned with data availability, discovery stage, and therapeutic objectives in AI-driven bioactive peptide discovery for metabolic diseases. Classical supervised learning algorithms are essential for initial screening tasks and ADMET prediction, especially when curated features such as cleavage motifs and physicochemical descriptors are accessible (Qiu et al., 2020; Tripathi et al., 2021). However, they possess inherent limitations in modeling the complex relationships among sequence, structure, and function that are characteristic of multi-target metabolic peptides.

Deep learning architectures, particularly transformer-based models, are increasingly effective in identifying subtle sequence determinants of bioactivity and facilitating *de novo* peptide design (Kensert et al., 2021; Li et al., 2022; Srivastava et al., 2024). However, their success depends on the availability of large, balanced, and biologically relevant training datasets, which are currently limited in the metabolic peptide domain. This gap can be addressed through transfer learning, utilizing large protein language models (ProtBERT, ESM) to create context-aware peptide embeddings that demonstrate strong generalization in few-shot learning contexts.

Ensemble and hybrid approaches represent a practical solution, integrating the interpretability and efficiency of classical models with the representational capabilities of deep networks (Li et al., 2022; Sharma et al., 2021). These architectures can significantly enhance candidate prioritization for next-generation biotherapeutics in the context of multi-objective optimization, including potency, protease stability, and non-toxicity.

Table 2 demonstrates that no singular modeling paradigm is universally superior. The future of AI-guided metabolic peptide discovery is characterized by integrative strategies, including transfer learning for embedding initialization, deep learning for complex pattern extraction, and ensemble modeling for robust decision-making. This directly supports the article's central thesis that multi-modal, AI-driven pipelines will transform the discovery of multifunctional peptides aimed at metabolic diseases.

10. Model training and validation strategies

The development of robust, reliable, and generalizable machine learning (ML) models for bioactive peptide discovery requires careful attention to model training and validation strategies. Given the complexity of biological systems and the high-dimensional nature of peptide data, it is essential to implement rigorous evaluation frameworks that not only assess predictive performance but also ensure reproducibility, interpretability, and biological plausibility.

10.1. Cross-validation techniques: ensuring robustness across data variability

Given the often-limited size and inherent imbalance in labeled peptide datasets, where active peptides are frequently outnumbered by inactive ones, cross-validation techniques play a critical role in mitigating overfitting and ensuring model robustness across different data splits.

One widely used method is K-Fold Cross-Validation, where the dataset is divided into k equal-sized folds. The model is trained on k-1 folds and validated on the remaining one, repeating this process k times (Ballester & Mitchell, 2010). This approach provides a balanced estimate of performance and is particularly valuable for moderately sized datasets. For imbalanced datasets, Stratified K-Fold Cross-Validation is preferred, as it preserves the class distribution across all folds (Kuh et al., 2024), reducing the risk of biased performance estimates.

In cases where datasets are extremely small, Leave-One-Out Cross-Validation (LOO) can be employed (Bizzotto et al., 2024; Kuh et al., 2024). Although computationally expensive, LOO minimizes bias by using each sample as a test case exactly once, making it suitable for high-quality datasets where every data point is critical. Additionally, Nested Cross-Validation is often used during hyperparameter tuning. This two-layer approach separates model selection from performance evaluation, preventing information leakage and ensuring unbiased assessments (Bizzotto et al., 2024; Kuh et al., 2024).

These cross-validation techniques are essential for ensuring that AI models generalize well to unseen data, rather than merely memorizing patterns in the training set. By addressing challenges such as data scarcity and class imbalance, they help build more reliable and interpretable models for peptide discovery.

10.2. Performance metrics: quantifying predictive accuracy and relevance

Bioactive peptide categorization depends on the use of strong performance criteria as false positives and negatives have major biological consequences (Fig. 4).

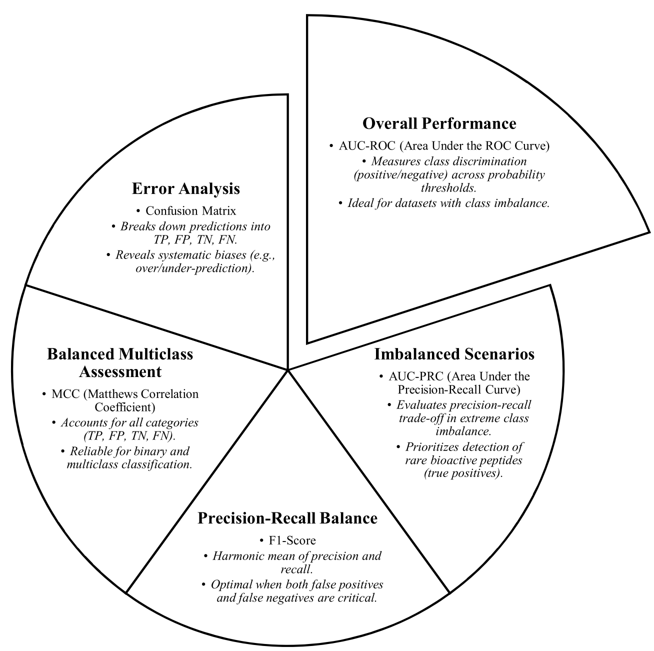

Fig. 4. Performanc metrics schema for bioactive peptide classification.
Legend: AUC-ROC: Area Under the Receiver Operating Characteristic Curve; AUC-PRC: Area Under the Precision-Recall Curve; F1-Score: Harmonic Mean of Precision and Recall; MCC: Matthews Correlation Coefficient; TP: True Positives; FP: False Positives.

While the AUC-ROC ranks total discriminative capability across thresholds, the AUC-PRC gives precision-recall trade-offs top priority in unbalanced data (Azad et al., 2024; Y. Zhang et al., 2024). Important for uncommon bioactive peptide identification, the F1-score harmonizes precision and recall; it also offers a fair evaluation for binary and multiclass tasks by including all confusion matrix categories. Confusion matrix analysis exposes systemic prediction biases, therefore allowing focused model improvement (Chen et al., 2024). These measurements used together guarantee thorough assessment of model dependability in challenging, high-stakes biological settings, transcending crude accuracy measures.

Assessing the predictive accuracy and relevance of AI models in bioactive peptide discovery requires careful selection of performance metrics, which are significantly influenced by the specific context. In addition to standard metrics like overall accuracy and R^2 for classification and regression tasks, a more refined evaluation necessitates the inclusion of metrics that address class imbalance and the varying costs associated with prediction errors. In the classification of bioactive versus inactive peptides, where inactive compounds generally outnumber active ones, metrics such as the Area Under the Receiver Operating Characteristic (ROC) curve (AUC) serve as a reliable indicator of a model's capacity to differentiate between classes across multiple thresholds, presenting a comprehensive perspective that is not reliant on a particular cut-off (FH et al., 2021; Kumar & Indrayan, 2011; Thieme et al., 2023). In addition to AUC, Precision, Recall (Sensitivity), and the F1-score are essential metrics. High recall is essential for identifying all potentially active peptides for subsequent experimental screening, prioritizing the reduction of false negatives, even at the cost of accepting some false positives (Abbas et al., 2025). Conversely, Precision becomes important when experimental validation is resource-intensive, rendering the reliability of positive predictions essential. The Matthews Correlation Coefficient (MCC) is a significant metric as it delivers a balanced evaluation, even in the presence of imbalanced classes, by considering true and false positives and negatives, thus providing a more accurate representation of a model's overall performance (Hogan et al., 2025; Li et al., 2024). In regression tasks, Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) measure prediction deviation, while R^2 indicates the proportion of variance explained by the model, together providing an assessment of the model's predictive capability (Dikmen et al., 2025). The primary metric is determined by the specific biological inquiry, the subsequent application, and the comparative costs linked to various prediction errors in the peptide discovery process.

10.3. Interpretability tools: bridging the gap between prediction and mechanism

Although deep learning models exhibit superior predictive capabilities, their opaque nature poses a considerable challenge in biomedical applications, where comprehending the rationale behind predictions is essential for directing experimental validation and formulating mechanistic hypotheses. In response to this limitation, various interpretability tools have been created.

One prevalent method is SHAP (SHapley Additive exPlanations), which utilizes cooperative game theory to allocate contribution values to each feature, such as amino acid positions or physicochemical properties, toward the final prediction (Akbar et al., 2022). This allows researchers to identify residues or motifs that influence bioactivity, thereby aiding in the rational design of peptides. LIME (Local Interpretable Model-agnostic Explanations) approximates complex models in a localized manner by employing simpler, interpretable models such as linear regression (Iqbal et al., 2025; Srivastava et al., 2024). LIME is especially useful for examining outliers or ambiguous cases, elucidating the model's decision-making process in particular situations.

Attention visualization serves as a significant tool in transformer-based architectures. Attention maps indicate the residues prioritized by the model during predictions, frequently emphasizing functional domains or binding interfaces within a peptide sequence (Bizzotto et al., 2024; Kensert et al., 2021; Srivastava et al., 2024). Saliency maps and gradient-based attribution methods in convolutional and recurrent neural networks pinpoint input regions, including specific amino acids, that significantly impact the output (Kensert et al., 2021; S. Zhang & Li, 2022). These techniques elucidate the structural determinants of bioactivity, connecting computational predictions to biological mechanisms.

The integration of interpretability tools enables researchers to enhance trust in AI-driven predictions and formulate testable hypotheses for subsequent experimental inquiry. This integration enhances the relationship between computational modeling and empirical research, thereby promoting the discovery and optimization of bioactive peptides.

10.4. Benchmarking across models and datasets: toward standardization and reproducibility

The use of AI models in peptide discovery is growing, and with it the necessity for regular benchmarking to evaluate approaches across tasks, datasets, and research teams. Benchmarking makes development quantifiable, reproducible, and driven by innovation rather than incremental modifications.

Effective benchmarking requires the utilization of standardized datasets, including BIOPEP-UWM, ChEMBL, or PeptideRanker, to facilitate fair and consistent comparisons (Bizzotto et al., 2024; Minkiewicz et al., 2019, 2022; Pearman et al., 2020; Srivastava et al., 2024). Moreover, consistent evaluation protocols, which encompass uniform cross-validation schemes, preprocessing steps, and performance metrics, are crucial for transparent assessments. To enhance reliability, reproducibility frameworks that promote open-source code, comprehensive documentation, and containerized environments facilitate independent verification of results. Structured initiatives such as CAFA (Critical Assessment of Functional Annotation) and community-driven challenges in peptide design are essential for offering controlled environments to assess model performance (Martin Alain, 2023; Saini et al., 2025). Establishing robust benchmarks enables the scientific community to progress beyond isolated demonstrations and promote cumulative knowledge-building. This collaborative initiative enhances the credibility of AI-based peptide discovery and expedites the application of these innovations in both research and clinical settings.

The achievement of standardization and reproducibility in AI-driven bioactive peptide discovery requires the establishment of stringent benchmarking protocols across various models and datasets. This involves a thorough evaluation framework that compares the predictive accuracies of different models (as detailed in Section 10.2) and examines their generalizability with external, unseen data (as outlined in Section 12.2). Recent studies utilizing comparative benchmarking methodologies have become essential in this field, offering analyses of the effects of various algorithmic architectures (e.g., traditional machine learning versus deep learning, as shown in Table 2), feature engineering strategies (Section 8), and data sources (as detailed in Table 1) on model robustness and their applicability to new peptide design challenges. The identification of best practices, clarification of the predictive capabilities of various AI approaches, and the expedited translation of *in silico* predictions into experimentally validated therapeutic candidates depend on these efforts. The establishment of standardized benchmarking frameworks, which include clearly defined metrics and publicly available datasets, is a critical area of research necessary for facilitating equitable and consistent comparisons within the swiftly changing landscape of AI tools in this field.

11. AI applications in predicting and designing bioactive peptides

11.1. Prediction of peptide bioactivity

One of the most impactful applications of artificial intelligence (AI) in bioactive peptide research lies in the prediction of functional properties from sequence data alone. This capability enables researchers to rapidly screen vast peptide libraries, whether derived from food proteins, endogenous precursors, or synthetic sources, and prioritize candidates with predicted therapeutic potential against metabolic diseases. AI models trained on curated datasets of experimentally validated peptides can identify patterns that correlate with specific biological activities, such as glucose regulation, appetite suppression, or inflammation modulation. These predictions not only accelerate discovery but also reduce reliance on laborious *in vitro* and *in vivo* assays.

11.1.1. Antidiabetic peptides: targeting GLP-1 and DPP-IV pathways

Type 2 diabetes mellitus (T2DM) continues to pose a significant global health challenge, characterized by insulin resistance and impaired β-cell function. In addressing this issue, glucagon-like peptide-1 (GLP-1) and its enzymatic regulator, dipeptidyl peptidase-4 (DPP-IV), have emerged as critical therapeutic targets (Flores-Hernández et al., 2024; Mamoudou et al., 2025; Sarker et al., 2024).

For instance, GLP-1 analogs offer promising avenues for treatment due to their ability to enhance glucose-dependent insulin secretion, suppress glucagon release, and slow gastric emptying. AI-driven approaches, leveraging sequence similarity, structural embeddings, and binding affinity predictions, have been instrumental in designing novel GLP-1 mimetics (Cai et al., 2017; Zhang et al., 2024). Deep learning models trained on known GLP-1 agonists and receptor interaction data can generate analogs with improved stability and potency, further refining their therapeutic potential.

Similarly, targeting DPP-IV, which rapidly degrades GLP-1, has proven effective in prolonging incretin activity and improving glycemic control. Supervised machine learning classifiers (Random Forest, Support Vector Machines) and deep neural networks have been widely used to predict DPP-IV inhibitory activity based on features like amino acid composition, *k*-mer frequencies, and physicochemical properties (Cai et al., 2017; Charoenkwan et al., 2022; Lu et al., 2024; Zhang et al., 2024). Notably, transformer-based language models like ProtBERT and ESM have shown exceptional performance by capturing evolutionary and structural signals relevant to enzyme inhibition (F. Zhang et al., 2024). These predictive tools have facilitated the discovery of bioactive peptides from diverse sources, including dairy, marine proteins, and plant-based hydrolysates, which exhibit potent antidiabetic effects with minimal side effects. By bridging computational innovation and biological insight, these advancements hold great promise for developing safer and more effective therapies for T2DM.

The prediction of antidiabetic peptides, especially those that target the glucagon-like peptide-1 (GLP-1) and dipeptidyl peptidase-IV (DPP-IV) pathways, has been enhanced through the use of traditional machine learning (ML) techniques, including Support Vector Machines (SVMs) based on physicochemical features, as well as advanced deep learning methods. Early studies predominantly utilized interpretable models such as Random Forest (RF) for classifying active sequences, due to their transparency and interpretability. Recent advancements in deep learning, particularly the emergence of transformer-based models (as outlined in Table 2), have shown an enhanced ability to capture intricate sequence-activity relationships from extensive, heterogeneous datasets. Deep learning models utilize extensive, pre-trained biological language models to deduce nuanced functional properties, frequently demonstrating enhanced generalization performance and predictive accuracy in the identification of novel active scaffolds, in contrast to traditional methods. Nonetheless, the opaque characteristics of these models require the creation and use of robust interpretability tools (Section 10.3) to clarify the underlying mechanisms of action, thus offering important insights into the connections between peptide sequences, structures, and biological activities.

11.1.2. Anti-obesity peptides: modulating appetite and energy balance

Obesity, a key component of metabolic syndrome, arises from chronic energy imbalance driven by dysregulated appetite, metabolism, and fat storage. Several endogenous peptides (leptin, ghrelin, peptide YY (PYY), and cholecystokinin (CCK)) play pivotal roles in regulating satiety and feeding behavior (Gangwal & Lavecchia, 2025; Rubinić et al., 2024).

For instance, leptin modulators have garnered significant attention due to leptin's role in signaling satiety to the hypothalamus (Hajfathalian et al., 2025; Marcone et al., 2017). However, leptin resistance, commonly observed in obesity, complicates its therapeutic application (Rubinić et al., 2024). AI-driven modeling has been employed to identify peptides that either mimic leptin's action or enhance leptin sensitivity through indirect mechanisms, such as gut hormone regulation. These approaches aim to restore balance in appetite control.

Similarly, appetite-suppressant peptides targeting PYY or CCK receptors show promise for weight management (Rubinić et al., 2024). Machine learning models trained on receptor-ligand interaction data can predict novel peptide sequences capable of inducing early satiety and reducing caloric intake. Convolutional neural networks (CNNs) have been used to identify short motifs associated with Y2 receptor activation, aiding the rational design of PYY3–36 analogs with enhanced efficacy (Gangwal & Lavecchia, 2025).

Moreover, reinforcement learning strategies are being explored to iteratively optimize peptides for improved receptor binding while minimizing off-target effects (Foluke Ekundayo, 2024; P. Peixoto & Azim, 2024). This approach is particularly valuable for designing orally stable and brain-penetrant appetite modulators, addressing challenges related to bioavailability and specificity. By integrating computational tools with biological insights, these innovations hold significant potential for developing effective therapies to combat obesity and its associated metabolic complications.

11.1.3. Anti-inflammatory peptides: targeting NF-κB and cytokine signaling

Chronic low-grade inflammation characterizes metabolic diseases and plays a role in insulin resistance, hepatic steatosis, and endothelial dysfunction. The nuclear factor kappa B (NF-κB) pathway is central to the inflammatory response and has been identified as a significant target for bioactive peptides (Hadipour et al., 2023; Wayal et al., 2025).

NF-κB inhibitors have been identified that suppress the activation of this pathway, consequently reducing the production of pro-inflammatory cytokines, including TNF-α, IL-6, and IL-1β (Karagiannis et al., 2024). Artificial intelligence models developed using molecular docking simulations and gene expression data have effectively predicted peptides that can disrupt IκB phosphorylation or the nuclear translocation of NF-κB subunits (Wu et al., 2024). Besides direct inhibition, certain peptides function as cytokine modulators, either promoting anti-inflammatory pathways or inhibiting pro-inflammatory pathways (Leutcha et al., 2025). Transformer-based models are highly effective in classifying peptides according to their immunomodulatory profiles, facilitating the identification of dual-action molecules that concurrently regulate multiple inflammatory pathways (Tripathi et al., 2021; Wu et al., 2024; Yetgin, 2025).

Numerous anti-inflammatory peptides originate from dietary sources, including milk caseins and soy proteins, as well as microbial metabolites, highlighting the significance of incorporating gut microbiome-derived sequences into AI-driven discovery processes (Mamoudou, Başaran, et al., 2024; Mamoudou & Mune, 2024; Marcone et al., 2017). Integrating computational innovation with biological insights offers significant potential for creating new therapies aimed at reducing inflammation in metabolic diseases.

11.2. De novo design and optimization of bioactive peptides

While the prediction of bioactivity from existing sequences represents a powerful application of artificial intelligence (AI), the next frontier lies in *de novo* peptide design, the generation of entirely novel sequences with tailored functional properties. This approach enables researchers to explore previously uncharted regions of the peptide sequence space, moving beyond natural templates to create optimized candidates with enhanced efficacy, stability, solubility, and pharmacokinetic profiles. Leveraging generative modeling, multi-objective optimization, and evolutionary algorithms, AI-driven *de novo* design is revolutionizing how we engineer peptides for metabolic disease prevention.

11.2.1. Generative models: learning the grammar of functional peptides

Generative models lie at the core of AI-driven *de novo* peptide design, enabling the creation of novel sequences that either mimic or surpass the functionality and druggability of known bioactive peptides. These models learn latent representations from large-scale datasets and generate new sequences that preserve key structural and functional motifs while introducing variability to enhance novelty and performance.

One prominent approach is Variational Autoencoders (VAEs), which encode peptide sequences into continuous latent spaces where similar sequences cluster based on shared physicochemical and functional traits (Xu et al., 2024). By sampling or interpolating within these clusters, researchers can generate peptides that balance similarity to known active sequences with structural diversity. VAEs have been successfully applied to design peptides with enhanced DPP-IV inhibitory activity and improved glucose-lowering effects, showcasing their utility in metabolic applications (Xu et al., 2024; Zhang et al., 2024).

Another powerful tool is Generative Adversarial Networks (GANs), which consist of two competing neural networks: a generator that creates new sequences and a discriminator that evaluates their authenticity against real data (Goles et al., 2024; Lin et al., 2022). Through iterative training, GANs produce high-quality synthetic peptides closely resembling experimentally validated ones.

Additionally, Reinforcement Learning (RL) frames peptide design as an optimization problem, where an agent iteratively mutates amino acid positions to maximize a reward function defined by desired properties such as binding affinity, ADMET profiles, or enzymatic resistance (Foluke Ekundayo, 2024; Peixoto & Azim, 2024). Techniques like Deep Q-learning and policy gradient methods have proven effective in optimizing peptides for receptor selectivity while minimizing off-target interactions, enabling precise targeting of complex biological pathways (Foluke Ekundayo, 2024; Peixoto & Azim, 2024).

Together, these generative approaches provide unparalleled flexibility in designing peptides with tailored biochemical signatures, paving the way for targeted interventions in complex metabolic diseases. By combining computational ingenuity with biological insights, they hold transformative potential for advancing peptide therapeutics.

11.2.2. Integration of ProtGPT2 in de novo peptide design

Expanding beyond conventional generative models, ProtGPT2, a sophisticated large language model for proteins, represents a significant paradigm shift in *de novo* peptide design by leveraging principles from natural language processing (Fan et al., 2025; Shah et al., 2024). Trained on a vast corpus of unlabeled protein sequences, ProtGPT2 learns the complex grammar, syntax, and statistical relationships inherent in protein evolution and structure-function (Fan et al., 2025; Shah et al., 2024). Unlike traditional generative adversarial networks (GANs) or variational autoencoders (VAEs) which often require explicit feature engineering or rely on fixed latent spaces, ProtGPT2 generates novel peptide sequences by predicting subsequent amino acids conditioned on preceding contexts, effectively 'writing' new peptides that adhere to known biological rules (Fan et al., 2025; Lai et al., 2025; Lin et al., 2022; Shah et al., 2024). This capability allows for the exploration of vast, unconstrained sequence spaces, identifying peptides with desired properties, from specific bioactivities to optimized physicochemical characteristics like stability and solubility, without direct experimental data for every design iteration.

The strength of ProtGPT2 lies in its capacity to generate highly diverse yet biologically plausible sequences, which can then be filtered and ranked based on *in silico* predictions (e.g., binding affinity via AlphaFold-Multimer) and subsequently synthesized for experimental validation (Fan et al., 2025; Shah et al., 2024). This deep exploration of sequence space, driven by evolutionary and structural patterns embedded within the model, positions ProtGPT2 as a transformative tool for accelerating the discovery of truly novel and functional peptide therapeutics.

11.2.3. Multi-objective optimization: balancing efficacy and druggability

Designing a bioactive peptide necessitates not only strong target engagement but also the careful consideration of various druggability criteria, including stability, solubility, absorption, and toxicity. Multi-objective optimization techniques tackle this challenge by incorporating various constraints into a cohesive scoring framework.

Efficacy is typically evaluated via predicted binding affinity, receptor activation potential, or modulation of downstream signaling (Leutcha et al., 2025; Mamoudou et al., 2025). Stability continues to be a significant challenge, especially as a result of peptide degradation by proteases. AI models forecast protease cleavage sites by analyzing sequence context and structural characteristics, facilitating the development of mutations that improve resistance. Solubility is essential for bioavailability, as inadequate aqueous solubility may restrict therapeutic efficacy. Predictive models utilize physicochemical descriptors, including charge distribution, hydrophobic moment, and isoelectric point, to inform advancements in this field (Andrusier et al., 2007; Daina et al., 2017).

Furthermore, ADMET properties are critical for clinical translation. Classifiers utilizing machine learning, trained on drug-like peptides and small molecules, assist in prioritizing candidates that exhibit advantageous pharmacokinetics and reduced side-effect risks (Bizzotto et al., 2024). These models guarantee that peptides demonstrate biological activity while fulfilling the practical criteria for therapeutic application.

To manage these competing objectives, multi-criteria decision-making frameworks, such as Pareto optimization, are employed (Kuh et al., 2024). These methods identify sequences that attain optimal trade-offs among all desired attributes, thereby circumventing the drawbacks of prioritizing one factor to the detriment of others. Systematic balancing of efficacy, stability, solubility, and ADMET properties facilitates the rational design of peptides, thereby enhancing their druggability and translational potential.

11.2.4. Evolutionary algorithms for sequence refinement

Evolutionary algorithms (EAs) replicate biological processes, including mutation, recombination, and selection, to optimize peptide sequences (Akbar et al., 2022; Singh et al., 2019). In this context, "fitness" is characterized as a composite score that encompasses both biological and physicochemical attributes.

Genetic Algorithms (GAs) function through the processes of crossover, mutation, and selection applied to a population of seed or randomly generated sequences across several generations (Fayaz et al., 2022; Moraes et al., 2017). Fitness functions typically include metrics like bioactivity, structural stability, and ADME predictions, directing the evolution of peptides toward favorable characteristics. Simulated Annealing (SA) utilizes a probabilistic approach that permits initial suboptimal mutations to avoid local optima, progressively converging to globally optimal solutions throughout the search process.

Alternative nature-inspired techniques, including Particle Swarm Optimization (PSO) and Ant Colony Optimization (ACO) (Herlambang et al., 2019; Mahmoodi-Reihani et al., 2020; Wu et al., 2024), demonstrate significant efficacy in navigating high-dimensional peptide landscapes. These algorithms are inspired by collective behaviors observed in nature, facilitating efficient navigation through extensive sequence spaces.

The integration of these strategies with deep learning models enhances their effectiveness by offering real-time feedback on the predicted attributes of each candidate. This synergy establishes a closed-loop design and evaluation system, facilitating the rapid identification of optimized peptides while addressing multiple objectives. These methods provide effective strategies for addressing the intricacies of peptide optimization in drug discovery.

11.2.5. Integration with experimental validation and iterative learning

The true power of AI-driven *de novo* design emerges when computational predictions are coupled with experimental validation and feedback loops. Once computationally designed peptides are synthesized and tested *in vitro* or *in vivo*, their performance data can be fed back into the model to refine future predictions, an approach known as active learning. This iterative cycle enhances model accuracy and adaptability, especially in capturing subtle nuances of peptide biology that may not be apparent from static sequence analysis alone.

Moreover, advances in high-throughput screening (HTS) (Roy, 2021) and deep mutational scanning (DMS) (Randall et al., 2024) enable rapid empirical testing of thousands of designed variants, providing rich datasets for training increasingly sophisticated models.

11.3. Structure-based peptide design

While sequence-based AI models have significantly advanced the discovery and design of bioactive peptides, understanding their three-dimensional (3D) structure and interaction dynamics with target proteins remains essential for optimizing potency, selectivity, and pharmacokinetics. Structure-based peptide design integrates computational structural biology, molecular modeling, and artificial intelligence (AI) to predict how peptides fold, bind, and function within complex biological environments. This approach enables rational engineering of peptides that interact specifically with metabolic disease-related targets such as G protein-coupled receptors (GPCRs), enzymes, ion channels, and nuclear hormone receptors.

11.3.1. Molecular dynamics simulations integrated with AI

Molecular dynamics (MD) simulations provide atomic-level insights into the conformational behavior of peptides over time, capturing dynamic processes such as folding, binding, and dissociation (Das et al., 2024; Leutcha et al., 2025; Raimi et al., 2024). However, MD simulations are computationally expensive and traditionally limited to short timescales or small systems.

Recent advancements have integrated machine learning and deep learning techniques to enhance the efficiency and accuracy of molecular dynamics simulations, thereby addressing existing limitations. Neural network potentials, including those created by DeepMind and SchNet, utilize quantum mechanical data to more accurately approximate energy landscapes compared to classical force fields (Abbas et al., 2024; Nussinov et al., 2022; Yetgin, 2025). AI-driven enhanced sampling methods, such as variational autoencoders (VAEs) and reinforcement learning (RL), direct simulations toward biologically relevant conformations, thereby decreasing computational costs and broadening the exploration of conformational space (Foluke Ekundayo, 2024; Goles et al., 2024; Xu et al., 2024). Moreover, the integration of metadynamics with machine learning-biased collective variables facilitates the precise exploration of rare events, including peptide-receptor associations and allosteric transitions.

Integrative approaches yield important insights into peptide flexibility, solvent exposure, and residue-residue interactions, which are crucial for the design of stable and active molecules aimed at metabolic pathways. Integrating traditional MD with AI-driven innovations enables researchers to address complex biological questions with enhanced efficiency and precision.

11.3.2. Docking and binding affinity prediction

Understanding how a peptide interacts with its molecular target at the atomic level is crucial for improving binding affinity, selectivity, and functional outcome. Computational docking algorithms predict the most favorable orientation and conformation of a peptide when bound to a protein receptor (Lei et al., 2021; Mamoudou et al., 2025; Nandwa et al., 2024; Raimi et al., 2024).

Artificial intelligence has transformed rigid-body and flexible docking through the enhancement of scoring functions and the optimization of search algorithms, resulting in improved accuracy and efficiency in predicting peptide-protein interactions (Esteva et al., 2021; Lei et al., 2021; Mullowney et al., 2023; Yang et al., 2023).

Deep learning-based scoring functions mitigate the limitations of traditional methods, which frequently fail to account for non-linear contributions to binding. Models such as Pafnucy, DeepDTA, and GraphDTA utilize graph-based representations of ligand-protein complexes to predict binding affinities with high accuracy (Lai et al., 2025; Thieme et al., 2023; Yang & Cheng, 2025). These methods enhance the reliability of docking predictions considerably.

Furthermore, peptide-specific docking tools like HADDOCK, Rosetta FlexPepDock, and AutoDock CrankPep have been augmented with AI components (Ballester & Mitchell, 2010; Fadahunsi et al., 2024). The upgrades enhance the tools' ability to account for the intrinsic flexibility and extended conformations of peptides, which are essential for accurately modeling their interactions with target proteins.

Recent advancements in end-to-end peptide docking models have been facilitated by transformer-based architectures that are trained on extensive interaction datasets (Clementel et al., 2025; Yang et al., 2023; Zhang et al., 2024). These models predict binding modes and affinities directly from sequence information, thereby removing the necessity for labor-intensive manual setup.

11.3.3. AlphaFold-based modeling of peptide-protein interactions

The advent of AlphaFold2, developed by DeepMind, has revolutionized structural biology by enabling highly accurate prediction of protein structures from amino acid sequences (Bryant et al., 2022; Nussinov et al., 2022; Yang et al., 2023). While initially optimized for globular proteins, recent extensions of AlphaFold and complementary tools like AlphaFold-Multimer have enabled the modeling of peptide-protein complexes, opening new avenues for structure-based drug design (Bryant et al., 2022; Nussinov et al., 2022; Yang et al., 2023).

The applications of AlphaFold2 in peptide research illustrate its versatility and significance. Predicting peptide binding modes has been notably enhanced by the adaptation of AlphaFold2 to model interactions between short peptides and the structured domains of larger proteins (Bryant et al., 2022; Nussinov et al., 2022; Yang et al., 2023). The integration of constraints derived from evolutionary covariation and geometric compatibility significantly improves the accuracy of predicted binding interfaces.

Furthermore, AlphaFold2 tackles the issue of modeling intrinsically disordered peptides, which frequently lack structure in isolation yet assume specific conformations upon binding, a phenomenon referred to as induced fit (Clementel et al., 2025; Nussinov et al., 2022; Yang et al., 2023). Specialized variants trained on disordered regions and co-evolutionary data yield more reliable predictions of dynamic transitions, enhancing understanding of their functional roles.

Moreover, AlphaFoldDB, a publicly available database of predicted protein structures, provides access to millions of modeled structures, including many involved in metabolic regulation, which can be used as docking partners or scaffolds for peptide engineering (Clementel et al., 2025).

11.3.4. Synergistic integration with experimental validation

Structure-based design attains practical significance solely when corroborated by experimental methods including X-ray crystallography, cryo-electron microscopy (cryo-EM), NMR spectroscopy, or surface plasmon resonance (SPR) (Acharya et al., 2024; Al-Baidhani et al., 2024; Minasov et al., 2022; Tang et al., 2024). AI models are progressively integrated into workflows to improve efficiency and precision. They facilitate mutagenesis experiments by predicting essential contact residues, prioritize candidate peptides for synthesis and testing according to predicted binding scores, and enhance low-resolution structural data through AI-assisted model fitting.

The incorporation of computational modeling with empirical validation enhances the iterative processes of design, testing, and refinement, thereby advancing AI-designed peptides toward therapeutic use. These approaches facilitate a connection between prediction and experimentation, enhancing the reliability and significance of peptide discovery.

11.4. Target identification and pathway analysis

Understanding the molecular mechanisms by which bioactive peptides exert their physiological effects is crucial for validating their therapeutic potential and guiding rational drug development. While many peptides are identified based on observed biological activity (such as glucose-lowering or anti-inflammatory effects) their specific molecular targets (such as G protein-coupled receptors (GPCRs), enzymes, ion channels, or nuclear receptors) are often unknown or poorly characterized. Artificial intelligence (AI) has arisen to bridge this knowledge gap by linking peptides to their likely molecular targets and situating them within broader signaling networks and disease pathways.

11.4.1. Use of AI to link peptides to molecular targets

Identifying the direct binding partners of bioactive peptides is complex, particularly when peptides display pleiotropic effects or operate via indirect mechanisms. AI-based approaches offer scalable and predictive solutions that enhance traditional experimental methods, including affinity chromatography, mass spectrometry, and CRISPR screens, to address this challenge (D. Das & Acharjee, 2024; Zhou et al., 2018).

A significant method involves sequence-based target prediction, utilizing deep learning models trained on extensive datasets of established ligand-receptor interactions to directly predict potential targets from peptide sequences (Imre et al., 2025; Kensert et al., 2021; Srivastava et al., 2024). Methods such as protein-ligand interaction fingerprinting, Siamese neural networks, and graph convolutional networks (GCNs) effectively model peptide and target features in a unified embedding space, facilitating high-throughput inference of target interactions (Kensert et al., 2021; Lei et al., 2024; Srivastava et al., 2024).

A significant tool pertains to peptide-protein interaction networks. Platforms like DeepDTA and GraphDTA utilize multimodal representations, integrating sequence embeddings, structural data, and pharmacophore descriptors to forecast peptide interactions with specific proteins (Lei et al., 2021, 2024; Yang & Cheng, 2025; Yetgin, 2025). These models have effectively identified new agonists and antagonists of metabolic receptors, such as GLP-1R, GIPR, and PPARγ, thereby advancing therapeutic discovery (Cai et al., 2017; Charoenkwan et al., 2022).

Furthermore, AI enhances cross-species and off-target prediction, supporting preclinical model selection through the extrapolation of target interactions across different species (Lu et al., 2023; Yang & Cheng, 2025). It also forecasts unintended interactions with other receptors or enzymes, which is essential for safety profiling and reducing off-target effects.

AI-driven capabilities allow researchers to formulate testable hypotheses regarding a peptide's mechanism of action, thereby facilitating downstream validation using methods such as site-directed mutagenesis, receptor knockout studies, or functional assays. The incorporation of computational predictions with experimental validation notably enhances the speed of discovering and optimizing bioactive peptides.

11.4.2. Network pharmacology and systems biology approaches

Given the multifactorial nature of metabolic diseases, single-target interventions often fall short in achieving sustained clinical benefits. This has spurred interest in network pharmacology, an approach that views diseases as perturbations of complex biological networks and seeks multi-target therapeutics that restore system-level homeostasis.

Artificial intelligence is essential in the construction and analysis of biological networks, providing a systems-level perspective on peptide activity (Esteva et al., 2021; Srivastava et al., 2024; Y. Yang & Cheng, 2025). Peptide-centric interaction networks are developed by synthesizing data from databases including STRING, Reactome, KEGG, and BioGRID (Simopoulos et al., 2020; Tran et al., 2018; Wu et al., 2024). These networks delineate direct receptor interactions and downstream components, such as signaling nodes, transcription factors, and effector molecules, thereby offering a comprehensive perspective on a peptide's impact.

AI utilizes centrality measures, such as degree, betweenness, and eigenvector centrality, to identify key network hubs based on this foundation (Simopoulos et al., 2020; Tran et al., 2018; Wu et al., 2024). These hubs frequently signify essential regulatory points that may function as innovative therapeutic targets or biomarkers. Moreover, AI-driven network analysis enhances multi-target drug repositioning by identifying synergistic peptide combinations or repurposing existing peptides for novel applications (Simopoulos et al., 2020; Tran et al., 2018; Q. Wu et al., 2024). A peptide originally designed for antihypertensive purposes may interact with insulin signaling pathways, indicating possible applications in diabetes management.

Viewing peptides as modulators of complex biological networks, rather than as isolated ligands, provides researchers with deeper insights into their systemic metabolic effects. This consolidative approach enhances the understanding of peptide biology and accelerates the discovery of novel therapeutic strategies.

11.4.3. Pathway enrichment analysis for mechanism elucidation

Beyond individual targets and networks, understanding the functional context of a peptide's activity requires placing it within relevant biological pathways. AI-enhanced tools streamline functional analysis through several key approaches (Gangwal & Lavecchia, 2024; Kar & Leszczynski, 2023; Yetgin, 2025). First, automated annotation leverages ontologies like Gene Ontology (GO), KEGG, and WikiPathways (Kar & Leszczynski, 2023) to assign peptides to biological processes such as glucose uptake, lipid metabolism, or inflammation. This ensures efficient and scalable classification of peptide functions.

Next, contextual pathway mapping employs AI models trained on transcriptomic and proteomic data from metabolic tissues, such as liver, adipose, and pancreas, to place peptides within disease-specific pathways (Erfanian et al., 2023; Kaczor-Urbanowicz & Wong, 2020). For instance, a peptide reducing NF-κB activation can be linked to inflammatory cascades in obesity-related insulin resistance, providing insights into its therapeutic relevance.

Finally, dynamic pathway simulation goes beyond static mapping by modeling how peptide interventions influence pathway behavior over time (Bryant et al., 2022; Clementel et al., 2025; Kar & Leszczynski, 2023). These simulations integrate kinetic models, dose-response data, and feedback loops to predict long-term effects under varying physiological conditions. These analyses offer mechanistic clarity, helping researchers determine whether a peptide acts through direct receptor modulation, enzyme inhibition, gene expression changes, or indirect effects mediated by the microbiota or immune system. Such insights are critical for advancing peptide-based therapies in metabolic diseases.

11.4.4. Integration with multi-omics and personalized medicine

The future of AI-driven target and pathway discovery hinges on its integration with multi-omics data to advance personalized peptide therapy. By leveraging patient-specific profiles, AI models can predict individual variability in target expression and responsiveness, identify subpopulations most likely to benefit from specific peptide interventions, and design combination therapies tailored to an individual's metabolic signature.

This systems-level, precision-oriented approach reflects the growing emphasis on stratified medicine in metabolic disease treatment. By moving away from one-size-fits-all solutions, it paves the way for targeted, mechanism-based therapies that address the unique biological and clinical needs of each patient. Such innovations hold immense potential to transform metabolic disease management and improve therapeutic outcomes.

11.4.5. Validation of AI-predicted peptides: transitioning from computational design to therapeutic development

The incorporation of artificial intelligence (AI) and machine learning (ML) into peptide discovery has revolutionized computational biology, facilitating the swift prediction of bioactive peptides with therapeutic applications. Deep neural networks (DNNs) and generative models, trained on large datasets of protein structures, binding affinities, and physicochemical properties, enable the design of peptides for various applications, including antimicrobial and anticancer therapies, as well as metabolic and dermatological interventions (Ali et al., 2024; Gangwal & Lavecchia, 2024; He et al., 2023; Saini et al., 2025).

While *in silico* predictions offer valuable initial insights, it is essential to conduct rigorous experimental validation to confirm efficacy, specificity, and safety. Initial screening generally utilizes *in vitro* assays, including surface plasmon resonance (SPR), enzyme-linked immunosorbent assays (ELISA), and circular dichroism (CD) spectroscopy, to quantify peptide-target interactions and evaluate structural integrity (Kelly et al., 2005; Musangi et al., 2024). AI-predicted antimicrobial peptides (AMPs) targeting Pseudomonas aeruginosa exhibit significant bactericidal activity in minimum inhibitory concentration (MIC) assays, supported by cytotoxicity profiling and fluorescence microscopy (Bae et al., 2025). AI-designed peptide inhibitors of the SARS-CoV-2 spike protein have demonstrated nanomolar binding affinity, confirming computational predictions (Silhan et al., 2023; Zdrazil et al., 2024).

The translation of these findings to preclinical models involves complexities including pharmacokinetics, biodistribution, and immune responses, thereby requiring comprehensive *in vivo* studies (Aissatou et al., 2025). In addition to infectious diseases and oncology, AI-derived peptides demonstrate potential applications in dermatology and metabolic disorders. Kennedy *et al.* (2020) identified pep_RTE62G, a natural peptide derived from Pisum sativum, through deep learning. Their findings demonstrate its anti-aging effects, which include enhanced fibroblast proliferation, increased extracellular matrix (ECM) synthesis, and clinical reduction of wrinkles. AI-predicted anti-diabetic peptides, such as pep_1E99R5, have been shown to significantly enhance glucose uptake in skeletal muscle cells and decrease fasting blood glucose levels in diabetic mouse models (Casey et al., 2021).

Despite these advancements, challenges remain, such as peptide stability, off-target effects, and scalability for clinical translation. Iterative AI-guided optimization, informed by omics data such as proteomics and transcriptomics, improves candidate refinement (He et al., 2023). Emerging strategies, such as the LIMITS workflow for the design of antitubercular host defense peptides (HDPs) from limited datasets, illustrate AI's adaptability in peptide discovery (Biswas et al., 2025).

Bridging AI-driven predictions with therapeutic success requires standardized protocols, interdisciplinary collaboration, and improvements in peptide delivery systems. AI is enhancing peptide discovery; however, systematic *in vitro* and *in vivo* validation is essential for translating computational advancements into clinically applicable precision medicines.

12. Challenges and limitations

Despite the remarkable progress made in applying artificial intelligence (AI) to the discovery and design of bioactive peptides for metabolic diseases, several significant challenges persist. These limitations span multiple dimensions, from data scarcity and model generalizability to biological complexity and ethical considerations. Addressing them is essential not only for advancing scientific understanding but also for ensuring that AI-derived discoveries translate effectively into real-world applications.

12.1. Data quality and availability

At the heart of any machine learning pipeline lies the data, and in the domain of bioactive peptide research, data availability remains one of the most pressing bottlenecks.

12.1.1. Fragmented and inconsistent datasets

Peptide bioactivity data can be found in many places, like scientific papers, patents, and databases such as BIOPEP-UWM, ChEMBL, and PeptideRanker (Davies et al., 2015; Minkiewicz et al., 2019, 2022). These datasets often have some limitations. First, they often lack clear annotations, with inconsistent or missing details about assay conditions, peptide concentrations, species origin, and target specificity. The data tends to focus on well-known peptides, leading to an overemphasis on certain bioactivities, like antihypertensive and antimicrobial effects, while other functions, such as insulin sensitization or lipid modulation, are less studied. Also, negative data, which is crucial for building strong and fair predictive models, is seldom recorded. This incomplete and unbalanced data makes it hard for AI algorithms to generalize well, which lowers their accuracy in predicting the bioactivity of new peptide sequences.

12.1.2. Sparse coverage of metabolic-relevant peptides

While large-scale protein databases such as UniProt contain extensive information on full-length proteins, they offer limited curated annotations specifically related to metabolically active peptides (UniProt Consortium, 2023). Endogenous peptides involved in glucose homeostasis, appetite regulation, or inflammation are often cryptic fragments buried within precursor proteins, making systematic extraction and labeling difficult.

Moreover, many peptides identified in peptidomics or food digestion studies remain functionally uncharacterized due to the lack of high-throughput assays linking sequence to specific metabolic effects.

12.1.3. Lack of standardized formats and ontologies

Not having standard formats and ontologies makes it harder to integrate data and build strong predictive models. Peptide sequences are often inconsistently represented, using formats like FASTA, IUPAC nomenclature, or annotated chemical modifications, and there's a lack of uniform standards. Activity labels are inconsistent, using terms like "antidiabetic," "insulinotropic," and "anti-inflammatory" interchangeably, even though they refer to different biological mechanisms. The inconsistencies make it hard for AI models to find useful patterns in varied inputs, which affects reproducibility and the chance for validation across studies.

12.1.4. Data bias and generalizability

The significant value of public peptide databases is undermined by a critical challenge, the pervasive issue of data bias, which directly affects the utility of AI models. Numerous prominent databases, including the Antimicrobial Peptide Database (APD) and subsets of BIOPEP-UWM, demonstrate a notable overrepresentation of specific peptide classes, such as antimicrobial, anticancer, and food-derived peptides, attributable to historical research priorities and the relative ease of experimental characterization (Bin Hafeez et al., 2021; Minkiewicz et al., 2019, 2022). The inherent bias in training data can significantly undermine the generalizability of AI models. Models primarily trained on biased datasets frequently exhibit a distorted comprehension of the peptide sequence-function relationship. They tend to excel in predicting properties for overrepresented classes but show inadequate predictive performance, or even systematic misclassification, when faced with underrepresented or genuinely novel peptide chemistries. This presents a significant analytical challenge: although models may demonstrate strong performance metrics on validation sets derived from the same biased distribution, their practical utility in identifying diverse bioactive peptides for metabolic diseases is considerably constrained. Addressing this requires strategic data curation to balance class distributions, the development of AI architectures resilient to imbalance, and rigorous external validation on unbiased datasets to accurately evaluate a model's capacity for generalized discovery.

12.2. Generalizability and overfitting

Even with high-quality data, AI models face the persistent challenge of generalization, the ability to perform reliably on unseen or out-of-distribution samples.

12.2.1. Model performance on external vs. training data

Many AI-based studies report impressive performance metrics on internal validation sets but fail to replicate results on external, independent datasets (M.K.G. Abbas et al., 2024; Saini et al., 2025). This discrepancy arises because:

- Training data are often drawn from a narrow range of sources, such as a single lab's experiments or a specific set of literature-derived entries.
- External data reflect different assay conditions, species models, or even slight variations in peptide synthesis methods, all of which can influence observed activity.

As a result, models may achieve high accuracy during training but exhibit poor predictive power when deployed in new contexts, limiting their utility in real-world discovery pipelines.

12.2.2. Domain shift and distributional mismatch

Another critical challenge is domain shift, which arises when the statistical characteristics of the training data differ from those encountered in real-world application settings. This gap affects how well the model can be applied in different situations and can hurt its accuracy in predictions.

One common issue is that models trained on human peptides often struggle with rodent data, and the opposite can happen too. Likewise, variations in tissues can change how peptides act; for instance, a peptide that works in the gut might have different effects in the liver or fat tissues. Additionally, variations in disease states can affect how well peptides work because of shifts in receptor levels, protease activity, or microbiome makeup.

Mismatches weaken AI predictions and increase the chances of false positives or negatives when models are used outside their intended context.

12.2.3. Overfitting to sequence motifs

Deep learning models, especially convolutional and transformer ones, are great at spotting complex patterns tied to bioactivity. This ability can turn into a problem when models get too attached to the quirks in the training data. For instance, they might pick up on patterns in synthetic peptides with unusual residues, focus on surface-level sequence traits instead of important biological interactions, or confuse sequence similarity with functional similarity, mislabeling structurally similar but inactive peptides.

To tackle overfitting, researchers often use techniques like regularization, dropout, cross-validation, and data augmentation (Ballester & Mitchell, 2010; Kuh et al., 2024). These techniques help make models stronger, but they can't completely fix the issues caused by a lack of diversity and representativeness in training data.

12.3. Biological complexity and multifactorial nature of metabolic diseases

Metabolic diseases such as obesity, type 2 diabetes mellitus (T2DM), non-alcoholic fatty liver disease (NAFLD), and metabolic syndrome are inherently complex, heterogeneous, and multifactorial in nature. Unlike monogenic disorders, these conditions arise from intricate interactions between genetic predisposition, epigenetic regulation, environmental influences, gut microbiota composition, and systemic inflammation. This complexity presents a major challenge for artificial intelligence (AI)-driven discovery pipelines aiming to identify bioactive peptides with therapeutic relevance.

12.3.1. Inter-individual variability and population generalizability of AI-predicted peptides

Due to human biological variety, AI-designed peptides in clinical practice are difficult to apply. Numerous studies have shown that genetic variations in metabolic regulators like FTO, PPARG, TCF7L2, and IRS1 can greatly affect peptide therapy responses, sometimes by several orders of magnitude (De Meyts, 2000; Mansoori et al., 2025; Sato et al., 2011). Beyond genetics, proteomic profiles, metabolic pathways, and microbiome composition interact to affect peptide absorption, distribution, and therapeutic effects (Lagier et al., 2012; Zivy & de Vienne, 2000). Recently conducted studies have measured numerous features of inter-individual variability. Peptide receptor and metabolizing enzyme genetic variants can vary binding affinities by 300 % and clearance rates by 200 % among demographic groups (Chuang et al., 2024). The gut microbiota significantly impacts peptide stability and bioavailability, with studies showing over fivefold changes in enzyme activity such as β-glucuronidase and proteases across people (Koliada et al., 2020). Dynamic epigenetic alterations including DNA methylation patterns and histone acetylation states add complexity, causing nutrition-sensitive peptide responsiveness changes (Zhang et al., 2020).

Several problems prevent current artificial intelligence systems from modeling this complex biological variety. Most notably, 78 % of genetic data in training datasets comes from European people. This imbalance ignores substantial ethnic differences in peptide metabolism and reaction. Current AI architectures also struggle to integrate multi-omics data streams and depict gene-environment interactions that affect therapy results.

These issues are being addressed by novel computational methods. Federated learning platforms offer secure analysis of decentralized datasets from over 300 healthcare institutions globally, protecting patient privacy and capturing population variety (Bertoni et al., 2017; Westerman et al., 2023). reinforcement learning algorithms use clinical responses from various patient groups to optimize peptide designs through adaptive feedback loops (Foluke Ekundayo, 2024; Peixoto & Azim, 2024). Most importantly, next-generation AI systems parallelly evaluate genomic, metabolomic, and microbiome information to better anticipate subgroup-specific treatment responses (He et al., 2023; Yetgin, 2025).

Despite these advances, AI peptide designs are still poorly translated to clinical practice. Fewer than 12 % of peptide treatment trials publish demographic subgroup outcomes, which is most problematic. It will need deliberate efforts to diversify clinical trial participation, extend biobank collections, and tighten treatment generalizability requirements across human groups to overcome this restriction. Overcoming these obstacles is essential to maximizing precision peptide medicine's promise.

12.3.2. Comorbidities and polypharmacology considerations

Metabolic disorders often come with other chronic issues like heart disease, high blood pressure, depression, and NASH, adding to the complexity of treatment (Acharya et al., 2024; Neuschäfer-Rube et al., 2023). In clinical practice, patients frequently show overlapping health issues that require multiple treatment approaches instead of just one medication.

Many bioactive peptides work on several biological pathways at the same time. GLP-1 analogs help control blood sugar, aid in weight loss, lower blood pressure, and reduce heart risks (Cai et al., 2017; Charoenkwan et al., 2022). These diverse effects are helpful in practice, but they make it harder to predict outcomes due to the added complexity in biology and mechanisms. AI models focused on single-target datasets, like DPP-IV inhibition, can miss off-target interactions, which might result in wrong predictions about effectiveness or side effects.

We need to rethink our approach and embrace multi-task and systems-level AI frameworks to tackle these challenges. These models need to combine disease networks, pathway interactions, and polypharmacological profiles to better capture the complexity of chronic diseases and improve peptide-based therapy development.

12.4. Experimental validation bottlenecks

Despite the increasing sophistication of AI-based discovery pipelines, the field continues to face a critical bottleneck: the gap between computational predictions and experimental validation.

12.4.1. Disconnection between in silico and wet-lab workflows

Through sequence analysis, docking simulations, or generative modeling, AI-driven investigations build large candidate peptide libraries. However, experimental confirmation of virtual predictions remains a major challenge. Limited peptide synthesis capacity (especially for long-chain or chemically modified sequences) and high expenses of *in vitro*, cell-based, and *in vivo* investigations are major drawbacks. From screening to biological validation, the experimental pathway might take months.

This disconnects leads to a situation where only a small fraction of computationally predicted peptides ever reaches experimental evaluation, reducing the feedback loop necessary for model refinement.

12.4.2. Preclinical testing constraints

Even with successful lab tests, moving peptide candidates to preclinical models still comes with challenges.

Differences in receptor structure and function between species can make it hard to apply lab findings to real-life situations, which lowers their reliability. Additionally, thorough pharmacokinetic profiling to assess absorption, distribution, metabolism, and excretion (ADME) involves significant animal testing, raising both resource concerns and ethical issues. Scalability is a big challenge right now. Current platforms just can't handle the volume needed to test many AI-generated peptides, which makes it hard to compare and optimize them effectively. These challenges slow down the shift from computational predictions to preclinical development, highlighting the need for better, ethical, and scalable validation processes.

To overcome these limitations, there is growing interest in automated wet-lab platforms, organ-on-a-chip technologies, and high-content screening assays that can accelerate validation while reducing reliance on traditional animal models.

12.5. Ethical and regulatory considerations

As AI becomes increasingly embedded in drug discovery and development, several ethical and regulatory concerns have emerged that demand careful consideration.

12.5.1. Transparency and explainability of AI models

AI models, especially deep learning architectures, are often criticized for their lack of interpretability, posing key concerns. Scientific reproducibility is hindered without transparent reporting of model architecture, training data, and parameters. Biological plausibility may be compromised if predictions do not align with established biochemical mechanisms. Moreover, clinical trust depends on the ability of stakeholders to understand and validate AI-driven decisions. Enhancing model transparency is thus essential for credibility and adoption.

The complexity of AI models, particularly those based on deep learning, raises concerns about reproducibility and reliability in AI-driven peptide discovery. The lack of transparency in model architecture, training data, and hyperparameters, as well as limited access to source code and trained model weights, hinders verification and advancement of research. To address this, promoting open science through public code sharing and adherence to standard reporting guidelines is essential. Transparency and reproducibility are crucial for establishing trust and advancing the responsible application of AI in biomedical research.

To address this, researchers are increasingly adopting explainable AI (XAI) techniques such as SHAP values, attention maps, and feature attribution methods to provide insights into model decision-making processes.

12.5.2. Intellectual property and data ownership

The emergence of AI-driven peptide discovery raises complex issues surrounding intellectual property and data ownership. Key questions include the patentability of AI-generated peptides and the attribution of rights, whether to the algorithm developer, platform user, or funding institution. Additionally, the use of proprietary or restricted datasets limits transparency and collaboration, while the inclusion of unpublished or unattributed data in training sets poses ethical concerns. Clear legal and ethical frameworks are urgently needed to balance innovation, intellectual contribution, and equitable access.

12.5.3. Regulatory frameworks for AI-derived therapeutics

Regulatory agencies such as the FDA and EMA are still establishing frameworks to assess AI-derived therapeutics, including peptides. Key challenges include defining validation standards for safety and efficacy, ensuring full traceability and auditability of the AI development pipeline, and implementing safeguards to monitor off-target effects and unintended outcomes. Initiatives like the FDA's Digital Health Pre-Cert Program and the EU's AI Act mark important progress, but comprehensive, science-driven policies are still needed to responsibly govern AI innovation in biomedical research.

13. Future perspectives and opportunities

As artificial intelligence (AI) continues to mature within the domain of bioactive peptide discovery, it is poised to catalyze transformative advancements across multiple dimensions, from personalized medicine and systems biology to synthetic design and clinical translation. The future of AI-driven peptide research lies not only in refining existing methodologies but also in expanding into novel frontiers that bridge computational modeling with biological complexity, ethical responsibility, and translational impact.

13.1. Integration with multi-omics and systems biology

The integration of AI with multi-omics technologies is advancing peptide research by facilitating a comprehensive understanding of their interactions within intricate biological networks. This method advances from fixed sequence-function relationships to dynamic, context-sensitive modeling of metabolic regulation.

The convergence enhances personalized medicine by facilitating the design of peptides customized to individual omics profiles, encompassing genetic background, microbiome composition, and disease state. AI models trained on gut microbiome data can predict individual responses to specific prebiotic or postbiotic peptides. Moreover, systems pharmacology improves therapeutic development by modeling peptide effects throughout metabolic pathways, thereby supporting multi-target strategies that align with the polygenic characteristics of metabolic diseases.

These advances signify a transition towards stratified, adaptive therapies that enhance efficacy, reduce side effects, and promote precision medicine.

13.2. Advances in explainable AI (XAI)

Despite their strong predictive performance, many AI models remain opaque, hindering their acceptance in high-stakes domains such as drug discovery and clinical decision-making. Explainable AI (XAI) aims to address this limitation by enhancing model transparency and interpretability (Alizadehsani et al., 2024; Jiménez-Luna et al., 2020).

Tools such as SHAP, LIME, and attention-based visualizations in transformer models allow researchers to identify the specific residues, motifs, or structural features that drive predictions. XAI also facilitates biological plausibility checks, ensuring that AI-generated insights align with established biochemical knowledge and avoiding misleading artifacts (Alizadehsani et al., 2024; Jiménez-Luna et al., 2020). Furthermore, as regulatory bodies increasingly require transparency, explainability is becoming essential for model validation and risk assessment.

By making AI outputs more interpretable and accountable, XAI fosters trust among researchers, clinicians, and regulators, accelerating the safe and responsible integration of AI into therapeutic development.

13.3. Collaborative platforms and open science initiatives

The advancement of AI-driven peptide discovery is contingent upon the availability of high-quality data, computational tools, and standardized evaluation frameworks. Recent initiatives promote collaboration and transparency via shared resources and community-led innovation.

Databases like BIOPEP-UWM, ChEMBL, and PeptideRanker are undergoing enhancements through the addition of more comprehensive annotations and standardized metadata. Harmonizing data formats and ontologies is crucial for enhancing interoperability among platforms. In addition, open-source AI tools have facilitated access to advanced models, enabling researchers to leverage existing frameworks and avoid redundant efforts.

Moreover, benchmarking competitions, inspired by initiatives such as CAFA (Bizzotto et al., 2024), promote reproducibility and thorough evaluation by encouraging the community to create and assess predictive models of peptide bioactivity.

These initiatives foster a collaborative and transparent research ecosystem, thereby accelerating innovation in peptide-based therapeutics.

13.4. Role of AI in microbiome-derived peptides: an emerging frontier

The human gut microbiome represents an extraordinary and largely untapped reservoir of bioactive peptides with profound potential for modulating host physiology and intervening in metabolic health (Hamadou, 2025; Ma & Ma, 2019; Simopoulos et al., 2020). AI is poised to revolutionize the discovery and characterization of these microbiome-derived peptides (MDPs) by navigating the immense complexity of metagenomic data and the intricate, often indirect, host-microbe interactions. Traditional reductionist approaches are insufficient to decipher this complexity (Ma & Ma, 2019; Simopoulos et al., 2020). Instead, AI-driven strategies, leveraging deep learning architectures and advanced graph-based models, are becoming indispensable. For instance, sophisticated sequence-based deep learning models (e.g., adapted transformer architectures or recurrent neural networks) can be trained on large-scale metagenomic sequencing data to predict novel peptide sequences from uncharacterized microbial genes, moving beyond homology-based identification. Concurrently, Graph Neural Networks (GNNs) can analyze intricate microbial interaction networks and their putative secreted factors, revealing how specific MDPs mediate cross-talk between microbial species and the host (Imre et al., 2025; Lei et al., 2024; Srivastava et al., 2024).

The true depth of AI's contribution lies in its capacity for functional inference from these novel MDPs. AI models trained on integrated multi-omics data (metagenomics, metatranscriptomics, metabolomics, and host phenomics) can decode complex microbiota-host signaling (X. He et al., 2023; Yetgin, 2025). This enables the prediction of MDPs that modulate appetite, inflammation, insulin sensitivity, and energy balance by inferring their likely interactions with host receptors, enzymes, or metabolic pathways (Hamadou, 2025). AI can predict novel postbiotic peptides derived from microbial fermentation, accelerating their identification and optimization for therapeutic relevance. Furthermore, AI facilitates the *de novo* design of engineered microbes capable of producing therapeutic peptides in situ within the gut environment, offering unprecedented precision-targeted delivery strategies. The challenge, and future opportunity, lies in robustly linking *in silico* predicted functions of novel MDPs to empirical *in vitro* and *in vivo* validation, necessitating integrated high-throughput screening and functional genomics. This profound integration of AI and microbiome research not only opens unprecedented avenues for developing microbiota-based peptide therapies but also positions the gut as both a source and a living biofactory for next-generation treatments, marking a transformative shift in personalized metabolic medicine.

13.5. Synthetic biology and AI-driven peptide libraries

Artificial intelligence is designing synthetic sequences for specific biological roles to improve peptide therapies, using machine learning and synthetic biology.

Variational autoencoders (VAEs), generative adversarial networks (GANs), and transformer-based architectures explore extensive sequence space to generate peptides with improved stability, target specificity, and pharmacokinetics. Furthermore, AI can improve non-natural changes such cyclization, stapling, and D-amino acid inclusion to increase proteolytic resistance and activity. AI-designed peptide libraries may be rapidly synthesized and tested using high-throughput screening tools, creating a feedback loop between *in silico* design and empirical validation.

This synergy between AI and synthetic biology heralds a new era of rational biodesign, where peptides are no longer limited by natural evolution but shaped by intelligent engineering.

13.6. Clinical translation pathways

The clinical translation of AI-driven peptide discovery requires coordinated efforts across multiple development phases to bridge computational innovation with therapeutic impact. To achieve this transition, four interconnected priorities must be addressed systematically. First, preclinical optimization demands integrated workflows combining computational ADMET prediction, organ-on-a-chip systems, and ethically optimized animal models, accelerating candidate selection while reducing reliance on conventional testing.

Central to this process is proactive regulatory engagement; establishing standardized evaluation frameworks for AI-designed molecules through early stakeholder dialogue will ensure rigorous assessment of data integrity, reproducibility, and safety profiles. Simultaneously, cross-sector collaborations between academic innovators, biotechnology firms, and pharmaceutical partners are imperative to advance translational pipelines through scaled manufacturing, adaptive clinical trial designs, and regulatory navigation.

Furthermore, post-approval monitoring strategies leveraging real-world evidence, including digital biomarker tracking and pharmacovigilance networks, will enable continuous refinement of AI algorithms based on longitudinal therapeutic outcomes. By prioritizing these synergistic approaches through multidisciplinary cooperation, the field can overcome existing translational barriers. This paradigm shift promises to accelerate the development of precision peptide therapeutics, ultimately improving clinical management of metabolic disorders through intelligently engineered solutions.

13.7. Advancements in peptide delivery and formulation optimization driven by AI

The clinical translation of therapeutic peptides faces significant challenges, including poor bioavailability, rapid enzymatic degradation, and restricted permeability across biological barriers. Artificial intelligence (AI) is transforming these fields by facilitating intelligent peptide formulation, chemical modification, and predictive optimization, thereby moving from traditional empirical methods to data-driven design. Machine learning (ML) and deep learning models, developed using comprehensive datasets of peptide physicochemical properties, stability profiles, permeability assays, and pharmacokinetic data, enable researchers to predict optimal delivery strategies prior to experimental validation (Al Hagbani et al., 2025; Saini et al., 2025; Yang & Cheng, 2025). AI enables the rational design of nanoparticle-based peptide formulations by predicting optimal combinations of polymers, lipids, and surfactants, while also optimizing nanoparticle properties including size, surface charge, encapsulation efficiency, and release kinetics (Al Hagbani et al., 2025; Saini et al., 2025; Yang & Cheng, 2025). AI-driven predictive modeling facilitates structural modifications, including PEGylation, cyclization, fatty acylation, and D-amino acid substitutions, by predicting their effects on protease resistance, biodistribution, and target specificity. This approach reduces off-target effects and enhances pharmacokinetics (Al Hagbani et al., 2025; Saini et al., 2025; Yang & Cheng, 2025).

In addition to formulation, AI is essential for predicting peptide interactions with biological barriers, thereby improving translocation across the intestinal epithelium, blood-brain barrier, or cellular membranes. Reinforcement learning algorithms have effectively optimized lyophilization conditions, excipient selection, and pH parameters to enhance peptide stability (Foluke Ekundayo, 2024; Peixoto & Azim, 2024). Advancements in AI-driven structural prediction tools, exemplified by AlphaFold, have significantly transformed the design of cell-penetrating peptides (CPPs) and peptide-drug conjugates through precise simulation of peptide-protein interactions and conformational dynamics (Sutcliffe et al., 2025). Recent advancements in generative adversarial networks (GANs) facilitate *de novo* peptide synthesis, allowing for the creation of sequences with customized absorption, distribution, metabolism, and excretion (ADME) profiles, thereby marking a significant development in precision medicine (Lai et al., 2025; Saini et al., 2025; Sutcliffe et al., 2025). Despite advancements, significant challenges remain, such as the standardization of training datasets, model interpretability, and the integration of AI with high-throughput experimentation (Dikmen et al., 2025; Yetgin, 2025). As AI methodologies evolve, their integration with multi-omics data and advanced drug delivery systems is set to transform peptide therapeutics, leading to next-generation formulations with enhanced clinical efficacy.

14. Conclusion

The convergence of Artificial Intelligence (AI) with bioactive peptide research marks a profound paradigm shift in the quest for novel therapeutics for metabolic diseases. This review examines how AI-driven methodologies are transforming all stages of discovery, including the acceleration of functionally relevant peptide identification, rational design, intelligent optimization, and mechanistic interpretation. The emergence of advanced deep learning models, such as sophisticated transformer architectures and generative AI, has advanced the field beyond traditional, labor-intensive experimental screening methods. Computational tools enable the *in-silico* assessment of numerous peptide candidates, significantly decreasing the time, cost, and resources typically required for drug development.

AI's significant potential extends beyond efficiency; it is fundamentally capable of modeling and interpreting biological complexity. Integrating multi-omics data, utilizing network pharmacology, and employing systems biology approaches allows for a comprehensive understanding of complex peptide-pathway interactions through AI. This systems-level understanding is crucial for addressing the polygenic and multifactorial characteristics of metabolic diseases, facilitating the development of precision-targeted interventions.

Despite significant advancements, achieving the full potential of AI requires addressing ongoing challenges. The challenges encompass the fragmentation and heterogeneity of biological data, the persistent demand for improved model interpretability, and the significant disparity between *in silico* predictions and thorough empirical validation. Moreover, the establishment of standardized benchmarks, the promotion of open-access data sharing, and the development of collaborative platforms are essential for ensuring reproducibility and facilitating cross-disciplinary knowledge exchange.

The future of AI in peptide discovery is fundamentally connected to strong interdisciplinary collaboration. Ongoing collaborations among computational biologists, clinicians, nutritionists, pharmaceutical scientists, and regulatory experts are essential for translating AI-generated insights into safe, effective, and scalable therapies. As ethical and regulatory frameworks develop, it is essential to prioritize transparency, accountability, and equitable access to AI-driven innovations.

AI is set to transform the field of metabolic disease therapeutics. It functions not only as an automation tool but also as a driver for intelligent design, deep personalization, and innovative scientific discovery. The integration of advanced machine intelligence with biological insight signifies a new phase in peptide-based medicine, where next-generation therapies are intelligently engineered for improved health outcomes.

Ethical statement - studies in humans and animals

Not applicable.

CRediT authorship contribution statement

Hamadou Mamoudou: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Investigation, Data curation, Conceptualization. **Martin Alain Mune Mune:** Writing – review & editing, Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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